

Exit Dynamics from Organic Certification in Quebec*

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Abstract

In Quebec, organic-certification exits remained persistently high at 5–7% annually over 2021–2025, while entries fell sharply from 10.8% to 3.4%. By 2025, exits exceeded entries for the first time, contracting the certified stock and reversing a decade of expansion. Using enterprise-level data spanning all product categories, this paper documents three life-cycle exit patterns from organic certification. First, the exit hazard follows an inverted-U shape, peaking around the third year. Second, more than 50% of exits occur within five years of certification. Third, enterprises certified in more product categories face lower exit risk. A calibrated dynamic model with heterogeneous time preferences and idiosyncratic pest shocks can account for the observed hump-shaped hazard through the differential exit of impatient versus patient producers, and a declining tail consistent with the selection of more patient producers over time. Policy counterfactuals indicate that an 80% pesticide tax raises ten-year survival from 81.0% to 89.3%, while a 1% productivity penalty in organic production lowers ten-year survival to 71.8%. These results suggest that long-term retention is shaped by behavioural heterogeneity in time preferences and exposure to ecological (pest) risk. The findings point toward policies that combine early transitional support with measures that reduce pest-related risks.

Keywords: Organic farming, adoption and retention, farm exit dynamics, pesticide taxation

JEL Classification: Q12, Q18, Q57, Q58

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1 Introduction

Despite a long-run rise in consumer demand for organic products,¹ favorable policies (Jaime et al. 2016), and price premiums, organic farming sectors in many countries share a persistent pattern: high exit rates from certification, concentrated in the first few years after entry. In Quebec, the annual exit rate holds between 5% and 7% over 2021–2025.² Combined with a collapse in the entry rate from 10.8% to 3.4%, this produces the first net contraction of the certified stock in 2025. In Ireland, Lapple (2010) documents high abandonment in the drystock sector, with farmers most likely to exit after the first subsidy contract expires; in Washington State, Brady et al. (2023) finds that exits concentrate in the first two years.

This paper asks what characterises the life-cycle pattern of exit from organic certification and whether a dynamic exit model with preference heterogeneity can rationalise it. Quebec provides a natural setting: with 8.4% of agricultural enterprises certified organic versus 1–4% in other provinces, it is Canada’s leader, yet despite strong institutional support,³ exit rates remain elevated during the first five years after certification. The analysis combines survival econometrics with a dynamic structural model to characterise the correlates and mechanisms of exit, and to evaluate alternative policy designs. We begin by constructing a dataset from the *Portail Bio Quebec* covering 2023 through 2025. The aggregate data reveal a fragile equilibrium: until 2024, new certifications largely offset persistent exits; in 2025, gross entry falls well below the exit flow, producing the first year of absolute sectoral contraction since 2012.⁴

Using Cox proportional hazard, complementary log-log (cloglog), and logit models, we characterise how the exit hazard varies with enterprise characteristics. Certification age is the strongest correlate: exit risk rises through the first 3.5 years (43 months) and then declines, a non-monotonic profile consistent with a liability of adolescence (Bruderl & Schussler 1990, Bruderl et al. 1992). Certifier matters: enterprises certified by *Ecocert Canada* exit less often than those under *Quebec-Vrai* or *Letis S.A.*. Product category matters too. Maple and livestock certifications require multi-year capital commitments—orchard development and breeding cycles—and therefore select forward-looking producers; both categories carry substantially lower exit risk. Wild-harvest producers, by contrast, face elevated exit hazard. Diversification adds a further protective margin: enterprises certified across more product categories exit less often.

These reduced-form averages don’t reveal why the hazard is hump-shaped. The Canadian farm sector comprises highly heterogeneous producers operating under a common policy environment (Weersink 2018), and different producer types may contribute unequally to exit risk at different durations. We therefore develop a dynamic model of the conventional-versus-organic choice in which producers differ in patience (Duflo et al. 2011, Falk et al. 2018) and face idiosyncratic pest shocks. The framework builds on Hopenhayn (1992), with exit probabilities derived from a standard discrete-choice micro-foundation (Rust 1987, McFadden 1974). We calibrate the model to the aggregate hazard and survival series, using exit rates from the *Portail Bio Quebec* and production-cost and yield data from the *Centre de Reference en Agriculture et Agroalimentaire du Quebec* (CRAAQ). Two preference types and a three-state Markov pest

¹Globally, organic food and drink retail sales rose from approximately €125 billion in 2021 to approximately €135 billion in 2022 (Willer et al. 2024). The Canadian organic retail market reached CAD 9 billion in 2023, an 11% increase since 2020; the market “held strong” despite recent economic turmoil (Canada Organic Trade Association 2024).

²Authors’ calculations using data from *Portail Bio Quebec*.

³Quebec’s support instrument is the *Prime-Vert* programme, which is front-loaded: it subsidises conversion to organic agriculture but not ongoing compliance, leaving producers exposed to output volatility once the transition window closes.

⁴Based on the annual totals reported by the *Portail Bio* dashboard back to 2012 (<https://portailbioquebec.info/>).

process reproduce the non-monotonic hazard: impatient producers exit rapidly under early pest shocks, while patient producers endure short-term yield losses because organic production yields higher long-run profit than its conventional counterpart. The early exit of impatient producers leaves a progressively more patient surviving population, and the aggregate hazard declines.

Building on this baseline, we evaluate three policy scenarios. A one-year post-certification subsidy of \$100/ha leaves ten-year survival nearly unchanged (81.3% vs. 81.0% at baseline); extending the same subsidy to ten years lifts it to 92.9%, but gains dissipate once support lapses. An 80% pesticide tax,⁵ by permanently raising the cost of the conventional alternative, achieves a ten-year survival of 89.3%. Finally, a 1% productivity loss from intensified pest pressure lowers ten-year survival to 71.8% and erodes even the patient subset that underpins long-run sector stability.

Related Literature. These results connect to several strands of the literature on farm exit, organic certification, and intertemporal decision-making in agriculture.

A first strand examines exit from organic certification and reversion to conventional agriculture directly. [Sahm et al. \(2013\)](#) review reversion from organic to conventional agriculture across countries, identifying economic pressures, regulatory burden, and social isolation as recurring factors. [Flaten et al. \(2010\)](#) survey 220 Norwegian farmers who deregistered between 2004 and 2007 and find that 61% reverted to conventional farming, while 21% continued organic practices without certification, showing that certification exit need not imply exit from agriculture. [Koesling et al. \(2012\)](#) similarly document how personal disappointment and lack of social acceptance interact with regulatory and economic pressures. In the United States, [Brady et al. \(2023\)](#) use a long panel of Washington State certification records and find that exits concentrate in the first two years, while [Läpple \(2010\)](#) examines abandonment in the Irish drystock sector and links exit to attitudes, market conditions, and the expiry of subsidy contracts. This literature establishes important empirical regularities surrounding organic certification exit and reversion, but we are not aware of a comparable enterprise-level dataset for Quebec or Canada. Our three-year panel (2023–2025), constructed from successive extractions of the *Portail Bio Québec* registry, provides the first enterprise-level dataset of organic certification dynamics in the Canadian context and complements the US evidence by documenting a structurally different institutional environment.

A second strand examines the broader determinants of farm entry, exit, adoption, and dis-adoption outside the specific context of organic certification. On the entry and adoption side, [Jaime et al. \(2016\)](#), [Khaledi et al. \(2010\)](#), and [Chen et al. \(2022\)](#) show how CAP⁶ subsidy pillars, transaction costs, and farm size shape agricultural adoption and entry in Sweden, Saskatchewan, and Canada, respectively. On the exit side, [Mishra et al. \(2014\)](#) and [Dong et al. \(2016\)](#) identify off-farm employment, government-payment intensity, technical efficiency, and scale as correlates of farm exit in the United States, while [An & Butler \(2012\)](#) apply a discrete-time duration framework to technology dis-adoption among California dairy producers. This broader literature establishes the importance of economic, institutional, and farm-level conditions in shaping agricultural entry and exit, but it does not examine the lifecycle of organic certification exit or the dynamic mechanism underlying its non-monotonic hazard. Our model fills this gap: two discount-factor types facing a Markov pest process generate a non-monotonic hazard profile

⁵This rate is calibrated to achieve the 12% pesticide-reduction target of Quebec’s *Sustainable Agriculture Plan* ([Séguin & Thiam 2025](#)).

⁶The Common Agricultural Policy (CAP) is the European Union’s framework of agricultural subsidies and rural development programmes.

and a compositional shift toward patient producers that reduced-form specifications cannot produce.

A third strand examines how time preferences and dynamic decision-making shape agricultural transitions. [Leblanc et al. \(2021\)](#) show that high discount rates among soybean producers in the Brazilian Amazon constrain the adoption of ecological practices, while [Piovesan \(2019\)](#) documents in a multiple-price-list experiment with Quebec producers that organic corn and soybean farmers exhibit higher risk tolerance than conventional farmers. On the modelling side, [Kuminoff & Wossink \(2010\)](#) develop an option-value model of conversion to organic farming under policy uncertainty, and [Pietola et al. \(2003\)](#) formulate the farm exit decision as a recursive optimal stopping problem. These contributions establish that intertemporal preferences matter for agricultural transitions, but none embeds heterogeneous patience in a dynamic model calibrated to administrative organic certification data. Our calibration recovers the patient/impatient composition of the Quebec organic sector and links it to the observed non-monotonic hazard profile, a quantitative object that, to our knowledge, the existing literature has not targeted.

Outline. The remainder of the paper is structured as follows. Section 2 introduces the data, describes the survival-analysis framework, and presents the empirical results. Section 3 develops the dynamic theoretical model. Section 4 calibrates the model and reports the policy simulations. Section 5 concludes.

2 Econometric Analysis

This section establishes the empirical facts that motivate the structural model. Subsection 2.1 describes the *Portail Bio Québec* panel and documents aggregate exit dynamics; Subsection 2.2 sets out the cloglog, logit, and Cox survival frameworks; Subsection 2.3 reports the estimates.

2.1 Data

Our data source is the *Portail Bio Québec* registry, which records the universe of enterprises currently certified organic in Quebec together with the entrants and exits over the preceding twelve months. We pulled the full registry in January 2024, February 2025 and February 2026, yielding three consecutive years of enterprise-level entry and exit records (2023, 2024 and 2025).⁷ All enterprise-level analyses in this paper use the 2023–2025 detailed panel.

Exit in the registry is an all-or-nothing enterprise-level event: no enterprise is observed dropping one certified product category while retaining others, so exit corresponds to the enterprise ceasing to be certified entirely rather than to a gradual downsizing of its product portfolio. Once an enterprise is no longer certified organic, the registry has no visibility into whether the producer continues to operate under conventional practices or has left the sector.⁸

⁷*Portail Bio* refreshes its website monthly and retains only the most recent twelve months of enterprise-level activity. We therefore pulled the snapshot once per year, each time capturing the complete twelve-month exit window before it was overwritten. For 2021 and 2022, only the aggregate *Portail Bio* dashboard is available, which reports annual totals back to 2012.

⁸[Flaten et al. \(2010\)](#), surveying 220 Norwegian farmers who deregistered between 2004 and 2007, find that 61% reverted to conventional farming while 21% continued organic practices without certification. [Brady et al. \(2023\)](#) documents that exiting Washington-State organic operations typically retain their land and equipment. In the absence of Quebec-specific post-exit data, we adopt the same baseline in the model of Section 3: certification exit corresponds to reversion to conventional farming.

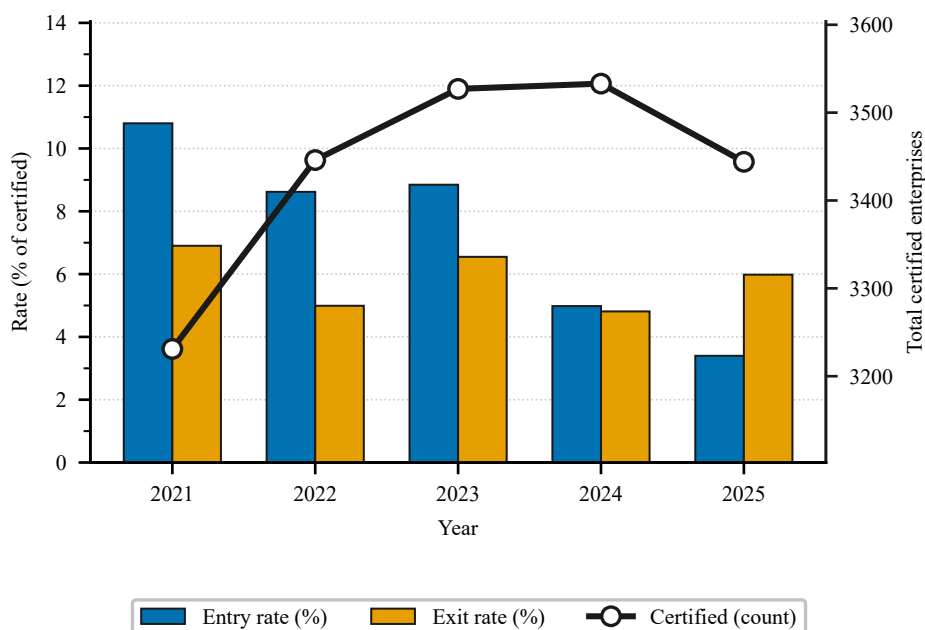


Figure 1: Evolution of Certified Organic Enterprises in Quebec, 2021–2025

Notes: This figure shows the evolution of the certified-organic stock in Quebec over 2021–2025. The black solid line reports the annual number of certified enterprises (right-hand vertical axis); the bars report new entries (blue) and exits (amber) as percentages of the previous year's certification stock (left-hand vertical axis). Aggregate counts for 2021–2022 are drawn from the *Portail Bio Québec* aggregate dashboard; 2023–2025 counts are computed from our enterprise-level detailed panel. *Source:* *Portail Bio Québec*, <https://portailbioquebec.info/>.

Aggregate evolution. The aggregate series in Figure 1 provides the broader context for the analysis. The recent contraction in certified enterprises follows from net entry turning negative in 2025, as the entry rate fell to 3.4% while the exit rate held near 6%; aggregate trends alone, however, do not reveal how exit risk evolves over the certification lifecycle. Between 2021 and 2024 the Quebec organic sector expanded on a thin margin: entry rates fell from 10.8% to 5.0% while the exit rate held between 5 and 7%, leaving the stock of certified enterprises to rise from 3,231 to 3,533. In 2025 the entry rate collapses further to 3.4% while exit climbs to 6.0%, and the certified stock contracts for the first time, falling by 89 enterprises to 3,444. Entry intensity in 2025 is less than one third of its 2021 level, while exit intensity remains within its 2021–2024 band.

Three disaggregated facts in Appendix A support a common-shock reading of the 2025 decline. First, the contraction is spatially uniform: exits exceeded entries in 13 of Quebec's 17 administrative regions in 2025, up from 6 in 2023. Second, it is compositionally uniform: every major product category except reconditioning registered a net outflow, and maple flipped from +11.3% of base in 2023 to –0.4% in 2025. Third, the shock concentrates on the entry margin: the ratio of one-product entries to one-product exits fell from 1.61 in 2023 to 0.63 in 2025, while the diversification profile of exits is unchanged.

These three patterns rule out compositional, regional, or size-based reallocation. The 2025 decline is a broad shock on the entry margin, and retention of already-certified producers becomes the lever for restoring growth. The analysis therefore turns to the timing and determinants of exit at the enterprise level.

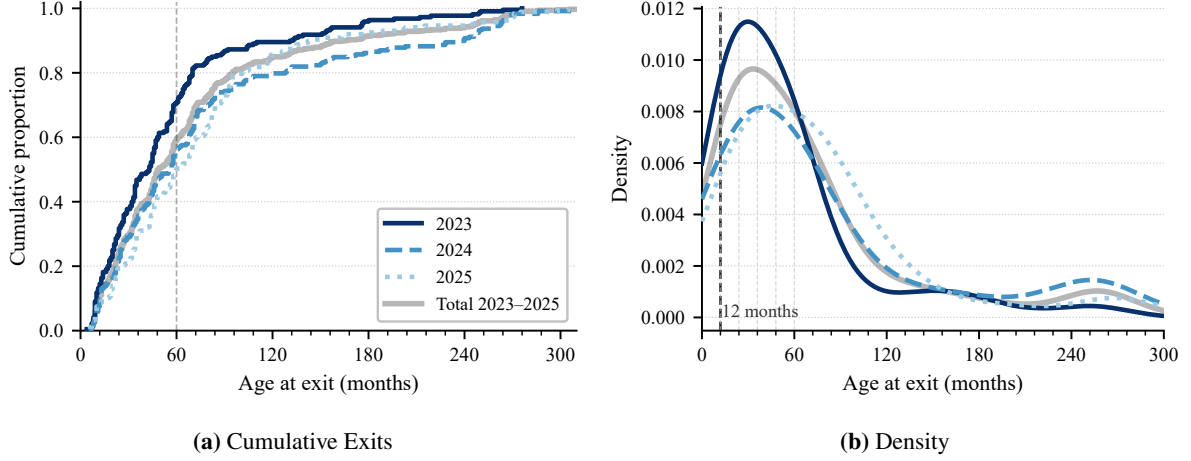


Figure 2: Distribution of Time Spent in the Organic Sector Before Exit

Notes: This figure shows the distribution of time spent in the organic sector, measured in months, by enterprises that exited in 2023 (solid dark navy), 2024 (dashed mid-blue), and 2025 (dotted light blue). The thick semi-transparent line shows the pooled 2023–2025 distribution. Panel (a) reports the empirical cumulative distribution function; panel (b) reports the kernel density estimate. The vertical marker in panel (b) highlights the 12-month anniversary of certification.

Age profile of exit. Exits are concentrated early in the certification spell. Figure 2 reports the cumulative distribution and kernel density of age at exit for the 2023, 2024, and 2025 exit cohorts. The three cohorts produce the same distribution. The density peaks between 24 and 48 months, and the cumulative distribution reaches 50% of exits by 60 months. This stability across cohorts indicates that the life-cycle exit pattern is a structural feature of organic certification, not a year-specific dynamic.

Appendix A reports the international and Canadian context, the structural composition of the sector by certifier, administrative region, and product category, and the disaggregated net change and entry–exit distribution figures that support the common-shock reading of the 2025 contraction above.

2.2 Econometric Specification

Our data exhibit two features that the estimator must account for. First, enterprises certified prior to 2023 enter the sample with spells already in progress at the start of the observation window (left truncation). Second, enterprises that remain certified at the end of 2025 contribute incomplete spells for which the exit time is unobserved (right censoring). These two features make a survival-analysis framework the natural choice (Cox 1972, Prentice & Gloeckler 1978, Jenkins 1995). Let T_i denote enterprise i ’s exit time, measured in months since first certification. The Cox proportional hazards model specifies the exit hazard at duration t as

$$\lambda_i(t \mid X_i) = \lambda_0(t) \exp(X_i' \alpha), \quad (2.1)$$

where $\lambda_0(t)$ is an unspecified baseline hazard at duration t and X_i includes production category, geographic region, and certifying body. To characterise how exit risk evolves with certification age, we complement the Cox model with a discrete-time hazard model using the complementary log-log (cloglog) link (Prentice & Gloeckler 1978, Jenkins 1995):

$$\log(-\log(1 - h_{it})) = \beta_1 \ln(t) + \beta_2 \ln^2(t) + X_i' \alpha, \quad (2.2)$$

where $h_{it} = \Pr(T_i = t \mid T_i \geq t)$ is the discrete-time exit hazard at duration t .⁹ A logit specification is retained as a robustness check.

Cox and cloglog absorb these two features through different mechanisms. The Cox partial likelihood (Cox 1972) restricts the risk set at each event time t to enterprises with $T_j \geq t$, so censored and left-truncated enterprises enter only the denominator and only when at risk. The cloglog binary likelihood absorbs them through the panel construction (Jenkins 1995): censored enterprises generate survival rows up to their last observed month with no terminal exit row, and left-truncated enterprises contribute rows only from the start of the observation window onward. Under non-informative censoring the two estimators are asymptotically equivalent (Prentice & Gloeckler 1978),¹⁰ so the coefficients in α retain their usual interpretation as proportional shifts in the conditional exit hazard.

Equation (2.2) yields two objects. First, the duration polynomial $\beta_1 \ln(t) + \beta_2 \ln^2(t)$ traces the exit life cycle but is subject to two confounders: cohort effects (different entry years imply different environments at a given age) and calendar-year shocks (common shocks affecting all certified enterprises at different ages). We address these by estimating specifications with entry-cohort fixed effects and, separately, calendar-year fixed effects (Table 1); they cannot be included jointly because age = calendar year – cohort. Second, the coefficient vector α reports conditional associations between enterprise characteristics and exit risk, with enterprise-clustered standard errors; region and certifier fixed effects absorb time-invariant differences in local conditions and certifier practices.

2.3 Results

Main regression results. Table 1 reports raw model coefficients from the cloglog and logit baseline specifications, along with log hazard ratios from the Cox proportional hazard model, all estimated on Equation (2.2).¹¹ Columns (1) and (4) present the cloglog and logit estimates controlling for entry-cohort fixed effects; columns (2) and (5) replace these with calendar-year fixed effects.¹² Column (3) reports log hazard ratios from the Cox model with cohort controls. Across all five specifications, the sign and significance of the key covariates are stable, lending confidence to the robustness of the findings.

The log-polynomial estimates confirm and quantify the inverted-U pattern in Figure 2. The positive coefficient on $\ln(\text{age})$ ($\hat{\beta}_1 \approx 2.04$, $p < 0.01$) combined with the negative coefficient on $\ln^2(\text{age})$ ($\hat{\beta}_2 \approx -0.27$, $p < 0.01$) implies that exit risk first rises then falls as certification age increases. Setting the first derivative of the log-hazard to zero gives a turning point at $\hat{t} = e^{\hat{\beta}_1/2\hat{\beta}_2} \approx 43$ months, or roughly 3.5 years after first certification. The non-monotonic hazard patterns documented above are the empirical objects that the structural model of Section 3 maps into latent producer types and pest-state dynamics, supplying the causal reading.

⁹Figure 2 shows that the density of age at exit peaks between 24 and 48 months, with most exits realised within the first five years; this non-monotonic shape motivates the log-polynomial specification for duration dependence used in Equation (2.2).

¹⁰Appendix B.1 illustrates the two data-construction structures (Tables B.1 and B.2).

¹¹The observation counts differ across specifications because Cox is estimated on the enterprise-spell dataset (one row per enterprise; see Table B.1) while cloglog and logit are estimated on the enterprise-month panel (one row per enterprise-month at risk; Table B.2). Appendix B.1 shows how the two likelihoods absorb right-censoring and left-truncation.

¹²The cohort grouping uses pre-2023, 2023, 2024, and 2025+ as bins. The two schemes cannot be included jointly because they are collinear with the log-polynomial duration term; the similarity of results across the two confirms that the findings are not driven by the choice of time control.

Table 1: Exit dynamics: raw model coefficients (cloglog and logit) and log hazard ratios (Cox)

	Cloglog		Cox	Logit	
	(1)	(2)	(3)	(4)	(5)
<i>Duration dependence</i>					
ln(age)	2.043*** (0.225)	1.281*** (0.169)		2.046*** (0.226)	1.285*** (0.170)
ln(age) ²	-0.271*** (0.032)	-0.215*** (0.025)		-0.271*** (0.032)	-0.216*** (0.025)
<i>Certifier (ref: Ecocert Canada)</i>					
Québec-Vrai	0.604*** (0.100)	0.806*** (0.097)	0.844*** (0.098)	0.604*** (0.100)	0.810*** (0.097)
Pro-Cert	0.130 (0.318)	0.423 (0.331)	0.373 (0.343)	0.130 (0.318)	0.425 (0.332)
Letis S.A.	1.059*** (0.121)	2.495*** (0.155)	3.085*** (0.353)	1.061*** (0.121)	2.506*** (0.155)
<i>Certified product type</i>					
Alcoholic beverages	-0.566* (0.312)	-0.513 (0.326)	-0.566* (0.322)	-0.567* (0.313)	-0.515 (0.327)
Maple syrup / forest products	-0.756*** (0.155)	-0.866*** (0.153)	-0.868*** (0.157)	-0.757*** (0.155)	-0.868*** (0.154)
Livestock	-0.703*** (0.178)	-0.273 (0.178)	-0.300 (0.186)	-0.704*** (0.178)	-0.274 (0.178)
Processed products	-0.583*** (0.132)	-0.233* (0.129)	-0.264** (0.133)	-0.584*** (0.132)	-0.233* (0.130)
Crop / plant products	-0.027 (0.137)	-0.051 (0.132)	-0.098 (0.134)	-0.027 (0.137)	-0.051 (0.132)
Organic baskets	-1.396* (0.731)	-1.218* (0.664)	-1.305* (0.694)	-1.398* (0.732)	-1.223* (0.665)
Wild harvest	0.583** (0.233)	0.671** (0.264)	0.670*** (0.258)	0.585** (0.233)	0.673** (0.265)
Observations	423,362	154,718	4,100	423,362	154,718
Region FE	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	No	Yes	Yes	No
Calendar-year FE	No	Yes	No	No	Yes

Notes: This table shows raw model coefficients in the cloglog and logit columns and log hazard ratios in the Cox column. Each certified product category enters as a binary indicator equal to one if the enterprise holds at least one certification in the category and zero otherwise. All specifications include region and certifier fixed effects. Columns (1) and (4) use entry-cohort fixed effects; columns (2) and (5) use calendar-year fixed effects; column (3) uses entry-cohort fixed effects. Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Certified product category. Each of the seven product-category regressors in Table 1 enters as a binary indicator for holding at least one certification in that category;¹³ each coefficient measures the effect of that category relative to enterprises without it, holding the other categories fixed. Maple/forest certifications carry the strongest protective effect (≈ -0.76 to -0.87 , $p < 0.01$), followed by livestock, processed products, and alcoholic beverages. Organic-basket enterprises show the largest point estimate (≈ -1.40 ,

¹³The seven product categories used throughout the paper are alcoholic beverages, crop/plant products, livestock, maple/forest products, organic baskets, processed products, and wild harvest.

$p < 0.10$) but with wide confidence intervals. Crop/plant certifications show no significant effect. Wild-harvest is the only category associated with elevated exit risk ($\approx +0.58$ to $+0.67$, corroborated by the Cox log hazard ratio of $+0.67$, $p < 0.01$).

Certifier. Survival rates differ significantly across certifying bodies. Relative to Ecocert Canada (the reference), enterprises certified by Québec-Vrai face higher exit risk (coefficient ≈ 0.60 , $p < 0.01$), while Pro-Cert shows no statistically significant difference. Letis S.A. shows the largest differential: the cloglog coefficient rises from 1.06 under cohort fixed effects to 2.50 under calendar-year fixed effects, and the Cox log hazard ratio reaches 3.09 ($p < 0.01$). Whether this reflects differences in certifier practices—the stringency of audits or the level of support provided—or compositional differences in which enterprises self-select into each certifier remains an open question.

Diversification across certified activities. The diversification margin is quantitatively associated with exit risk. Replacing the binary product indicators with the count of distinct certified product categories (an enterprise scores one per category in which it holds at least one certified product, regardless of how many products it holds within that category) yields a precisely estimated negative marginal association: each additional category is associated with a 0.0005 lower monthly exit hazard in the cloglog and logit specifications ($p < 0.01$), as reported in columns (1), (3) and (5) of Table 2.

Table 2: Exit probability and number of certified product categories

	Cloglog (AME)		Cox (log HR)		Logit (AME)	
	(1)	(2)	(3)	(4)	(5)	(6)
Number of certified categories	-0.0005*** (0.0001)		-0.1397 (0.0974)		-0.0005*** (0.0001)	
Number of certified categories (excl. wild harvest)		-0.0006*** (0.0001)		-0.2249** (0.0996)		-0.0006*** (0.0001)
Observations	423,362	423,362	4,100	4,100	423,362	423,362
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Certifier FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Calendar-year FE	No	No	No	No	No	No

Notes: This table shows average marginal effects (AME) in the cloglog and logit columns and log hazard ratios in the Cox columns. Columns (1), (3) and (5) use the count of distinct certified product categories (the number of the seven product categories in which the enterprise holds at least one certification). Columns (2), (4) and (6) restrict the count to the six categories whose individual product indicators in Table 1 do not carry a significantly positive exit coefficient (alcohol, maple/forest, livestock, processed, crop-plant, organic baskets), excluding wild harvest. This restricted count isolates the diversification effect to purely protective product categories. All specifications include region, certifier, and entry-cohort fixed effects. Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Since wild harvest is the single category whose indicator enters the exit hazard with a significantly positive coefficient, the all-seven-category count mixes protective activities with one risk-increasing activity. Columns (2), (4) and (6) redefine the count to exclude wild harvest (the count now runs over the six protective categories only) while keeping the estimation sample, duration controls and fixed effects identical to columns (1), (3) and (5); the reported observation counts are therefore the same by construction, and the difference across columns reflects only the reweighting induced by the change of regressor. Isolating diversification across purely protective product categories in this way, the marginal effect strengthens to 0.0006 in the cloglog and logit specifications, and the Cox estimate sharpens from

a statistically insignificant -0.140 to a significant log hazard ratio of -0.225 ($p < 0.05$). The gradient is therefore not driven by how the count is constructed and holds across all three estimators.¹⁴

The pattern is consistent with a within-enterprise insurance mechanism: an enterprise certified across several activity categories can absorb category-specific disturbances through unaffected revenue streams, whereas a single-product enterprise must exit if its one activity fails. This mechanism also aligns with recent evidence in the broader agricultural-diversification literature: Adam & Abdulai (2024) document that crop diversification among Ghanaian smallholders reduces risk exposure, with the magnitude of the risk-reducing effect varying non-linearly with farm size, crop types, and market access.

By-product heterogeneity reported in Table B.3 (Appendix B) shows that the protective effect is strongest for alcohol and livestock enterprises and monotone for crop-plant producers; maple/forest and organic basket subsamples are omitted because the cloglog model does not converge within those near-zero-exit categories. Figure B.1 plots the predicted monthly exit probability against the number of certified product categories and illustrates the same negative gradient in levels.

Robustness. The main findings are stable across alternative specifications. First, we re-estimate the baseline using raw category counts (Table B.6 in Appendix B) and grouped counts collapsed into indicators for zero, one, or two or more products per category (Table B.7). Second, we restrict the sample to new entrants who obtained their first certification within the observation window to address potential left-truncation bias (Table B.8); the inverted-U age pattern is preserved and the maple/forest protective effect persists, but estimates for wild harvest and organic baskets are inestimable because of near-zero exit cells in this smaller subsample (662 enterprises). Third, we replace the log-polynomial duration controls with step-function age-band indicators (Table B.9). The age-band coefficients trace the inverted-U non-parametrically: they peak at $+0.0012$ over 37–60 months and decline to $+0.0005$ at 61+ months, consistent with the 43-month turning point implied by the log-polynomial estimates. Fourth, we decompose the diversification margin into a breadth component (number of distinct certified product categories) and a depth component (extra products within each already-certified category) and enter the two jointly (Table B.4). Across all these alternatives, the sign, magnitude, and significance of the product-type and diversification effects remain essentially unchanged.

Protection operates on the breadth margin on average: the breadth coefficient matches the column-(2) estimate of Table 2, while the aggregate depth coefficient is statistically indistinguishable from zero across all three estimators. The aggregate depth null masks meaningful category-level heterogeneity, however. Moving from one product to two or more within the same category adds further protection in maple/forest, livestock, and processed products (Table B.7 in Appendix B), where the “2+” coefficient is at least as negative as the “1” coefficient and significant at the 1–5% level; it does not add protection in crop/plant, and is irrelevant in wild harvest once enterprises have passed the single-product threshold. The aggregate scalar averages these opposing returns to zero. The overall pattern is consistent with the within-enterprise insurance mechanism: broadening across activities hedges independent, category-specific shocks, while deepening within an activity pays off only in the categories whose underlying activity is already protective.

¹⁴Appendix B.4 shows that this restricted-count coefficient is unchanged when wild-harvest status is entered as an explicit additional control, ruling out confounding from the omitted wild-harvest indicator.

3 Theoretical Framework

To rationalise the non-monotonic hazard profile and the early-exit concentration documented in Section 2, we develop a dynamic exit model with heterogeneous discount factors and idiosyncratic pest shocks. The econometric estimates of Section 2 describe associational gradients rather than causal effects; this model supplies the causal interpretation, extending the firm-exit framework of [Hopenhayn \(1992\)](#) and the discrete dynamic programming approach of [Rust \(1987\)](#) to the institutional setting of organic certification.

3.1 Heterogeneous Producers and Production Regimes

Producers differ in their intertemporal preferences, and this heterogeneity is the behavioural margin on which the model operates. [Falk et al. \(2018\)](#), using the Global Preference Survey across 76 countries and 80,000 respondents, find that within-country heterogeneity in patience exceeds between-country heterogeneity and correlates with savings and investment decisions. In the agricultural context specifically, [Duflo et al. \(2011\)](#) show that a non-trivial share of Kenyan maize producers are present-biased and underinvest in high-return fertilizer inputs, while a minority of time-consistent producers respond to small post-harvest nudges. These results motivate the two-type specification below. Each producer is indexed by a discount factor β_θ , with $\theta \in \{\underline{\theta}, \bar{\theta}\}$, where $\underline{\theta}$ represents an impatient type and $\bar{\theta}$ a patient type, such that $\beta_{\underline{\theta}} < \beta_{\bar{\theta}}$. While both types face the same expected profit opportunities, patient producers place greater weight on future income streams and are therefore more willing to stay in organic practices that may generate delayed but higher returns.

Production can occur under two regimes: conventional farming, C , and organic farming, O . These regimes differ in their input composition, exposure to environmental shocks, and the nature of market and policy incentives. Conventional farming relies on chemical inputs such as pesticides to stabilize yields, with pesticide use subject to regulatory taxation. In contrast, organic farming abstains from synthetic inputs, receives price premiums and per-ha subsidies, but remains exposed to biological risks such as pest infestations. In the conventional regime, production depends on the use of conventional inputs z_c and pesticide applications x . Per-period profits are defined as:

$$\pi_c(z_c, x) = p_c \cdot f(z_c, x) - \mu_z \cdot z_c - (1 + \tau)p_x \cdot x, \quad (3.1)$$

where $f(\cdot)$ denotes the production function, p_c is the output price, μ_z the unit cost of conventional inputs, p_x the pesticide price, and τ a pesticide tax rate. The use of pesticides mitigates pest-related losses. In the organic regime, producers face biological risk but benefit from policy support and market premiums. The per-period profit is:

$$\pi_o(z_o, \sigma^t) = (p_o + s) \cdot \sigma^t f(z_o) - \mu_o \cdot z_o - c, \quad (3.2)$$

where $p_o > p_c$ represents the organic price premium, s is a per-unit subsidy, z_o denotes organic-specific inputs, μ_o is their unit cost, and c is a per-hectare certification fee paid each period while certified. The production function $f(z_o)$ is the pest-free maximum output: realised output equals $\sigma^t f(z_o)$, with $\sigma^t \in (0, 1]$ a retained-yield fraction and $\sigma_1 = 1$ normalised to the no-shock benchmark. The stochastic term $\sigma^t \in \{\sigma_1, \dots, \sigma_N\}$ captures pest idiosyncratic shocks in the organic sector during period t , evolving

according to a Markov process with transition probabilities:

$$\Omega_{ij} = \text{Prob}(\sigma^{t+1} = \sigma_j \mid \sigma^t = \sigma_i), \quad i, j \in \{1, 2, \dots, N\}. \quad (3.3)$$

Each state σ_i represents a level of pest pressure affecting yields, ordered so that $\sigma_1 > \sigma_2 > \dots > \sigma_N > 0$, with $\sigma_1 = 1$ the pest-free benchmark (full retained yield) and σ_N the most adverse; the transition matrix Ω governs the dynamics of the process. Including these shocks captures realistic yield variability, consistent with the documented yield gap between organic and conventional farming (e.g., [de Ponti et al. 2012](#), [Ponisio et al. 2015](#)). The probability that the pest environment is in state σ_j after t periods of organic farming is denoted by $\Phi_{j,t}$, which evolves as:

$$\Phi_{j,t} = \sum_{i=1}^N \Omega_{ij} \Phi_{i,t-1}, \quad (3.4)$$

starting from a uniform initial distribution $\Phi_{j,0} = 1/N$ for all j . This assumption implies that upon entry into organic farming, pest conditions are initially unknown and equally likely across all possible states.

3.2 Dynamic Optimization

The exit decision rests on a comparison of two continuation values: what the producer earns by remaining in organic production, and what they would earn by switching permanently to conventional farming. Exit occurs when the conventional continuation value exceeds the organic one. The intertemporal discount factor $\beta_\theta \in (0, 1)$ governs how much weight each producer places on future payoffs relative to current profits.

Conventional continuation value. All policy and market variables are collected in the exogenous state vector $\Psi = (\tau, s, p_c, p_o)$, comprising the pesticide tax, organic subsidy, and output prices. The conventional value function satisfies the Bellman equation:

$$V_C^\theta(\Psi) = \max_{z_c, x} \left\{ \pi_c(z_c, x) + \beta_\theta V_C^\theta(\Psi') \right\}, \quad (3.5)$$

where the maximization is over conventional input choices and Ψ' denotes next-period conditions. This value is the producer's outside option: what they can secure by exiting organic farming.

Organic continuation value. The continuation value of organic farming for a producer of type θ in pest state σ^t is the present discounted value of remaining organic indefinitely, accounting for the stochastic evolution of pest conditions:

$$V_O^\theta(\Psi, \sigma^t) = \max_{z_o} \left\{ \pi_o(z_o, \sigma^t) + \beta_\theta \mathbb{E}_{\sigma^t} [V_O^\theta(\Psi', \sigma^{t+1})] \right\}, \quad (3.6)$$

where the expectation is taken over next-period pest states given the current state σ^t . The first term is current profit; the second aggregates future organic profits, discounted by β_θ and weighted by the Markov transition probabilities.

Optimal Input Choices and Value Functions. Each Bellman equation separates into a static input-choice problem and a dynamic continuation value. In the conventional regime, the first-order conditions give optimal inputs z_c^* and x^* :

$$p_c f_z(z_c^*, x^*) = \mu_z, \quad (3.7)$$

$$p_c f_x(z_c^*, x^*) = (1 + \tau) p_x. \quad (3.8)$$

Substituting into the stationary Bellman equation yields the closed-form conventional value:

$$V_C^\theta = \frac{1}{1 - \beta_\theta} \pi_c^*, \quad (3.9)$$

where $\pi_c^* \equiv \pi_c(z_c^*, x^*)$ is the maximized flow profit. A higher pesticide tax τ raises the cost of x , reduces x^* , and therefore lowers V_C^θ .

In the organic regime, the optimal input $z_o^*(\sigma_j)$ in pest state σ_j satisfies:

$$(p_o + s) \sigma_j f'(z_o^*) = \mu_o, \quad (3.10)$$

with maximized flow profit $\pi_o^*(\sigma_j) \equiv \pi_o(z_o^*(\sigma_j), \sigma_j)$. Collecting the N state-specific value functions into a vector $\mathbf{V}_O^\theta = [V_O^\theta(\sigma_1), \dots, V_O^\theta(\sigma_N)]^\top$ and flow profits into $\boldsymbol{\pi}_o^* = [\pi_o^*(\sigma_1), \dots, \pi_o^*(\sigma_N)]^\top$, the system of Bellman equations across pest states solves to:

$$\mathbf{V}_O^\theta = (\mathbb{I}_N - \beta_\theta \Omega)^{-1} \boldsymbol{\pi}_o^*, \quad (3.11)$$

where $\mathbb{I}_N$ is the $N \times N$ identity matrix. A unique solution exists for all $\beta_\theta < 1$ since the spectral radius of $\beta_\theta \Omega$ is strictly below one. The matrix inverse aggregates current losses and expected future recovery into a single state-contingent value. Section 4 evaluates Equation (3.11) numerically for calibrated parameter values.

The role of patience is already visible from these expressions. From Equation (3.9), V_C^θ rises with β_θ at rate $\pi_c^*/(1 - \beta_\theta)^2$. From Equation (3.11), $V_O^\theta(\sigma_j)$ also rises with β_θ , but the rate depends on the full distribution of future pest states through Ω . When organic farming is viable in expectation, that is, when π_o^* is large in favorable states, patience raises the organic value faster than the conventional value, making exit less likely. The formal statement of this comparative static follows after the exit probability is defined.

Exit Decision. A producer exits when $V_C^\theta(\Psi) > V_O^\theta(\Psi, \sigma_j)$: the continuation value of conventional farming exceeds that of remaining organic in the current pest state (Aguirregabiria & Mira 2010). In a deterministic model this generates a sharp threshold, a critical pest level $\sigma^*(\theta)$ below which every producer of type θ exits and above which none do, implying an exit rate of exactly zero or one within every type-pest-state cell. The observed hazard $\hat{P}_t$, by contrast, is an interior probability at every duration in the data, never zero or one, so matching it requires exit to be probabilistic rather than deterministic. Producers of the same type θ facing the same pest state σ_j still differ in unobserved switching costs, so exit is modeled as a probability rather than a binary indicator. Following Rust (1987) and McFadden

(1974), treating those unobserved costs as iid extreme-value noise yields the logit exit probability:

$$P_{t,\sigma_j}^\theta = \frac{1}{1 + \exp[-\gamma(V_C^\theta(\Psi) - V_O^\theta(\Psi, \sigma_j))]}, \quad (3.12)$$

where $\gamma > 0$ governs how sharply exit responds to the value gap. As $\gamma \rightarrow \infty$ the logit collapses to the deterministic threshold rule; as $\gamma \rightarrow 0$ exit is driven by idiosyncratic noise. The calibration in Section 4 estimates γ alongside the other structural parameters by matching the model's predicted exit rate to observed data.

The probability that a producer of type θ exits the organic sector after t periods, across all pest shock states, is given by:

$$P_t^\theta = \sum_{j=1}^N \Phi_{j,t} P_{t,\sigma_j}^\theta, \quad (3.13)$$

Any perturbation that shifts probability mass toward higher pest states; either through an unfavorable realization or a change in the transition matrix Ω , translates into a higher aggregate exit probability. As the continuation value of remaining in organic production, $V_O^\theta(\Psi, \sigma^t)$, rises with σ^t , adverse pest realizations (low σ^t) lower it and thereby raise the conditional exit probability P_{t,σ_j}^θ via Equation (3.12). Thus, pest shocks influence exits through two reinforcing channels: they increase the probability of being in unfavorable pest states, $\Phi_{j,t}$, and raise the propensity to exit within those states, P_{t,σ_j}^θ . Both effects push upward the overall likelihood of exit, P_t^θ . The magnitude of these effects depend on the producer's type θ .

The following proposition formalizes the mechanism emphasized in the model. In the model, impatient producers are more likely to exit under pest pressure because they discount future recovery more heavily. Patient producers internalize the long-run value of certified status and sustain participation even when current shocks depress organic profits.

Proposition 1 (Patience and exit). *Suppose the calibrated fundamentals $(\pi_o^*, \pi_c^*, \Omega)$ satisfy*

- (i) $\pi_o^*(\sigma_N) < \pi_c^*$: *the most adverse pest state yields lower organic profit than conventional, so exit is an economically meaningful choice;*
- (ii) $\omega^\top \pi_o^* > \pi_c^*$, *where ω is the stationary distribution of Ω : the long-run mean of organic profit exceeds conventional.*

Then, the aggregate exit probability satisfies: $P_t^\theta > \bar{P}_t^\theta$ for every $t \geq 1$. Equivalently, P_t^θ is strictly decreasing in β_θ at the calibrated $(\hat{\Omega}, \hat{\pi}_o^, \hat{\pi}_c^*)$.*

Proof. The sign of $\partial P_{t,\sigma_j}^\theta / \partial \beta_\theta$ coincides with the sign of $\partial V_C^\theta / \partial \beta_\theta - \partial V_O^\theta(\sigma_j) / \partial \beta_\theta$. Differentiating the closed forms of V_C^θ and $V_O^\theta(\sigma_j)$ and expanding via Neumann series reduces the sign question to that of

$$G(\sigma_j, \beta_\theta) \equiv \sum_{t=1}^{\infty} t \beta_\theta^{t-1} \left\{ \mathbb{E}_{\sigma_j} [\pi_o^*(\sigma^t)] - \pi_c^* \right\}.$$

Under (i), early-horizon $\delta_t(\sigma_j) \equiv \mathbb{E}_{\sigma_j} [\pi_o^*(\sigma^t)] - \pi_c^*$ can be negative when the chain is likely to visit σ_N ; under (ii), late-horizon $\delta_t(\sigma_j)$ converges to a strictly positive value. Whether the positive tail dominates

in the weighted sum is a quantitative question whose sign is verified at the calibrated primitives.¹⁵ Hence $\partial V_O^\theta(\sigma_j)/\partial \beta_\theta > \partial V_C^\theta/\partial \beta_\theta$, so $\partial P_{t,\sigma_j}^\theta/\partial \beta_\theta < 0$ for every pest state. The aggregate $P_t^\theta = \sum_j \Phi_{j,t} P_{t,\sigma_j}^\theta$ is a non-negative weighted sum with β_θ -invariant weights, so $\partial P_t^\theta/\partial \beta_\theta < 0$. ■

Survival Dynamics. Let S_t^θ denote the share of the original type- θ entry cohort, normalized so that $\sum_\theta S_0^\theta = 1$, that remains certified organic at the start of period t . The evolution of survival follows the recursive process:

$$S_t^\theta = S_{t-1}^\theta (1 - P_{t-1}^\theta), \quad (3.14)$$

where the initial shares S_0^θ represent the population composition at certification, and are normalized such that $\sum_{\theta \in \{\underline{\theta}, \bar{\theta}\}} S_0^\theta = 1$. Aggregate exit flows from the organic sector in period t are $E_t = \sum_{\theta \in \{\underline{\theta}, \bar{\theta}\}} S_t^\theta P_t^\theta$, which, using Equation (3.13), can be rewritten as:

$$E_t = \sum_{\theta \in \{\underline{\theta}, \bar{\theta}\}} \sum_{j=1}^N S_t^\theta \Phi_{j,t} P_{t,\sigma_j}^\theta. \quad (3.15)$$

The conditional aggregate probability of exit is then $P_t = E_t/S_t$: the probability that a producer exits during period t , conditional on having remained certified through period $t - 1$, where $S_t = \sum_\theta S_t^\theta$ is the share of the original entry cohort still certified at the start of period t .

Equations (3.14)–(3.15) map the model’s latent state space onto two observables: the share of surviving organic farms S_t and the period- t conditional exit rate P_t . These are the objects Section 4 matches to the data, using calibrated parameter values to conduct counterfactual policy simulations.

4 Quantitative Analysis

We calibrate the model to match the aggregate hazard and survival series recovered non-parametrically from the *Portail Bio Québec* panel. This disciplines the structural interpretation of the empirical patterns and provides the platform for the counterfactual policy simulations that follow. The analysis proceeds in three steps: calibration of production and behavioral parameters, evaluation of model fit, and examination of policy interventions under alternative scenarios.

4.1 Data and Calibration

The calibration identifies the structural parameters of the model describing the exit dynamics of organic and conventional producers. It relies on two complementary data sources: exit information from the *Portail Bio Québec* and production-cost and yield data for organic and conventional field crops from the *Centre de Référence en Agriculture et Agroalimentaire du Québec* (CRAAQ).

Production Parameters. The profit functions π_c and $\pi_o(\sigma^t)$ in Equations (3.1) and (3.2) are parameterised from CRAAQ average budget data on corn, soybean, and barley.¹⁶ CRAAQ reports only optimised

¹⁵At the calibrated primitives $(\hat{\Omega}, \hat{\pi}_o^*, \hat{\pi}_c^*)$ and both calibrated discount factors $\hat{\beta}_\theta$ and $\hat{\beta}_{\bar{\theta}}$, direct evaluation gives $G(\sigma_j, \beta_\theta) > 0$ for every pest state σ_j ; see Appendix C, Table C.1, for the full derivation and numerical values.

¹⁶CRAAQ publishes matched organic/conventional enterprise budgets only for field crops; no equivalent paired cost data exists for maple, livestock, or processed products. Within the set of products for which paired budgets exist, field crops also

outcomes per hectare, namely yield and total production cost for each regime, matched across the three crops (which together account for approximately 77% of certified field-crop acreage in Quebec between 2019 and 2022); it does not separately report the input quantities (z_c, z_o, x) or input prices (μ_z, μ_o, p_x) . We therefore cannot solve the first-order conditions (3.7)–(3.10); we instead take the maximised flow profits directly from the aggregated data, which equals the maximised profit by construction. Let $\bar{y}_c, \bar{C}_c$ denote the three-crop benchmark values of conventional yield and total cost, computed as the simple (unweighted) arithmetic averages across corn, soybean, and barley, and $\bar{y}_o, \bar{C}_o$ the corresponding organic quantities computed in the same way; the per-crop values are reported in Table D.1 (Appendix D).

CRAAQ reference budgets are prospective planning data constructed from enterprise accounting and validated by technical panels; they describe the well-managed representative farm and do not absorb realised pest or climate losses. The organic benchmark $\bar{y}_o$ is therefore identified with the pest-free maximum output $f(z_o^*)$ in Equation (3.2), evaluated at the optimal input z_o^* .¹⁷ At baseline (no pesticide tax and no organic subsidy), the maximised conventional flow profit is $\pi_c^* = p_c \bar{y}_c - \bar{C}_c$, and the maximised organic flow profit in pest state σ_j is

$$\pi_o^*(\sigma_j) = p_o \sigma_j \bar{y}_o - \bar{C}_o - c, \quad (4.1)$$

where $c = \$20/\text{ha}$ is the annual certification fee¹⁸ and $\sigma_j \in (0, 1]$ is the calibrated retained-yield fraction ($\sigma_1 = 1$ normalised; σ_2 and σ_3 with $\sigma_2 > \sigma_3$ recovered within the meta-analytic bounds of Seufert et al. (2012) reported in Table D.2).

Empirical Calibration Targets. The data cover exits observed through 2025; enterprises still certified at the end of 2025 are treated as right-censored survivors. The dataset is reorganised into a duration format: each enterprise contributes one record per year since certification, and duration t denotes years elapsed since first certification. We calibrate the model on the crop-plant subsample ($n_0 = 1,683, 365$ observed exits) for two reasons. First, crop-plant is the only category for which CRAAQ publishes matched organic/conventional enterprise budgets; the model’s production parameters $\pi_o^*(\sigma_j)$ and π_c^* are therefore identified specifically for this subsample. Second, Figure A.11 in Appendix A shows that the crop-plant age-at-exit density is the closest among all categories to the pooled 2023–2025 distribution, so the duration profile identified on crop-plant carries directly to the aggregate. The raw observed hazard at duration t is $\hat{P}_t = d_t/n_t$, where d_t is the number of crop-plant enterprises exiting at duration t and n_t is the corresponding risk set, enterprises that have neither exited nor been censored before duration t . The Kaplan–Meier product-limit estimator (Kaplan & Meier 1958) then recovers the empirical survival function:

$$\hat{S}_t = \prod_{s=1}^t (1 - \hat{P}_{s-1}), \quad (4.2)$$

offer the cleanest identification channel: the organic/conventional profit gap is dominated by the pesticide margin, since the conventional producer optimises over x while the organic producer forgoes it entirely.

¹⁷The planning-data interpretation is corroborated by the near-parity between organic and conventional benchmark yields in Table D.1 (organic within roughly 10% of conventional for corn and soybean). Realised farm-level organic yields in adverse Quebec years fall well below this gap; a cross-sectional mean that absorbed pest losses would not display near-parity.

¹⁸The \$20/ha figure is the annual per-hectare certification fee typical of Quebec organic crop operations, averaged across the main certifiers (Ecocert Canada, Québec-Vrai, Pro-Cert). Fees vary with farm size and certifier; for mid-sized crop operations the per-hectare charge typically falls in the \$15–\$25 range.

which accounts for right-censoring and is the exact empirical counterpart to the model's aggregate survival $S_t = \sum_{\theta} S_t^{\theta}$ in Equation (3.14). These two series, the period-by-period hazard $\hat{P}_t$ and the cumulative survival $\hat{S}_t$, are the calibration targets. Exit flows E_t carry no independent information: the model's S_t is entirely determined by $\{P_s\}_{s \leq t}$ through Equation (3.14), so targeting E_t separately would double-count the same moment.

Parameter Vector and Objective Function. Given a policy environment $\Psi = (\tau, s, p_c, p_o)$, the continuation values V_C^{θ} and V_O^{θ} follow from Equations (3.9) and (3.11). The state-contingent exit probability P_{t, σ_j}^{θ} is then computed from Equation (3.12), and the aggregate hazard P_t and survival S_t from Equations (3.13)–(3.15). The parameter vector to be identified is: $\chi = \{\beta_{\underline{\theta}}, \beta_{\bar{\theta}}, \gamma, S_0^{\theta}, \{\sigma_i\}_{i=1}^N, \Omega\}$, where $\beta_{\underline{\theta}}$ and $\beta_{\bar{\theta}}$ are the discount factors of impatient and patient producers, γ the logit sensitivity parameter, S_0^{θ} the initial share of impatient types, σ_i the pest-state productivity multipliers with $\sigma_1 = 1$ normalised, and Ω the Markov transition matrix.¹⁹ After the normalisation $\sigma_1 = 1$, the row-sum constraints on Ω , and the tridiagonal restriction, χ contains ten free parameters: two discount factors, the logit sensitivity, the initial share S_0^{θ} , two pest-state multipliers, and four independent entries of Ω . These are disciplined by twenty annual hazard targets (and the associated survival series, which is a deterministic transform of the hazard), so the calibration is over-identified at roughly a 2:1 ratio. These parameters are identified by minimising the weighted distance between model-implied and empirical moments:

$$\min_{\chi} \sum_{t=1}^T (P_t(\chi) - \hat{P}_t)^2 + \lambda_S (S_t(\chi) - \hat{S}_t)^2, \quad (4.3)$$

where $P_t(\chi)$ and $S_t(\chi)$ are the model-implied hazard P_t and survival S_t evaluated at χ , $T = 20$ years, and $\lambda_S = 100$.²⁰ The two moments carry complementary information: $\hat{P}_t$ captures the period-by-period shape of the exit hazard, while $\hat{S}_t$ anchors the cumulative retention level.

We minimise the objective in Equation (4.3) by Sequential Quadratic Programming, initiated from multiple starting values over the admissible space (see Table D.2) to avoid local minima. At each iteration, the parameter vector is updated, the type-specific survival trajectories are recomputed from Equation (3.14), and the simulated series $\{P_t, S_t\}$ are regenerated until convergence.

4.2 Results and Model Fit

The calibrated parameters are reported in Table 3. The two discount factors are $\hat{\beta}_{\underline{\theta}} = 0.90$ and $\hat{\beta}_{\bar{\theta}} = 0.97$. The initial share of impatient enterprises ($\hat{S}_0^{\theta} = 0.22$) implies that roughly one in five entrants is present-biased. The logit sensitivity $\hat{\gamma} = 0.002$ governs the sharpness of exit responses to profit differentials; the pest-state multipliers $\hat{\sigma}_2 = 0.90$ and $\hat{\sigma}_3 = 0.55$ indicate that severe shocks reduce retained yields by

¹⁹The pest process is approximated by a three-state Markov chain ($N = 3$), with σ_2 and σ_3 determined endogenously by the calibration. Following the agronomic evidence on gradual pest dynamics, Ω is restricted to a tridiagonal (banded) structure with $\Omega_{13} = \Omega_{31} = 0$, ruling out direct transitions between the most favourable and most adverse states in a single period; Table D.2 in Appendix D reports the literature anchors. The initial distribution $\{\Phi_{j,0}\}$ is assumed uniform across all states.

²⁰The truncation at $T = 20$ years is motivated by the instability of $\hat{P}_t$ beyond that horizon and by a shift toward demographic exit mechanisms outside the model (Figure D.1 in Appendix D); 94% of observed exits occur within twenty years, so truncation removes little information attributable to the modelled mechanism. The value $\lambda_S = 100$ reflects that Kaplan–Meier survival is estimated more precisely than the annual hazard (Kaplan & Meier 1958), and the calibrated $(\beta_{\underline{\theta}}, \beta_{\bar{\theta}}, \gamma)$ are stable across $\lambda_S \in \{25, 50, 100, 200\}$.

45%.

Table 3: Calibrated Parameters

Parameter	Description	Value
β_θ	Impatient discount factor	0.90
$\beta_{\bar{\theta}}$	Patient discount factor	0.97
γ	Logit sensitivity $(\$/\text{ha})^{-1}$	0.002
S_0^θ	Initial share of impatient enterprises	0.22
<i>Retained yield σ_i, by pest state</i>		
σ_1	Good state (no shock)	1.00
σ_2	Moderate shock	0.90
σ_3	Severe shock	0.55
<i>Transition matrix $\hat{\Omega}$ (rows = origin state)</i>		
$\hat{\Omega} = \begin{pmatrix} 0.152 & 0.848 & 0 \\ 0.199 & 0.220 & 0.581 \\ 0 & 0.148 & 0.852 \end{pmatrix}$		

Notes: This table reports the calibrated values of the model parameters, estimated by solving the minimization problem defined in Equation (4.3). Robustness to the literature-anchored bounds in Table D.2 is reported in Appendix D.2.

Time Preferences. The patient discount factor $\hat{\beta}_{\bar{\theta}}$ is identified strictly away from its calibration bounds, while the impatient factor $\hat{\beta}_\theta$ reaches its upper bound.²¹

To compare with the literature, which reports per-period discount rates rather than discount factors, we map the calibrated β_θ into the implied annual rate $r_\theta = 1/\beta_\theta - 1$. The impatient factor $\hat{\beta}_\theta = 0.90$ implies an annual discount rate of roughly 11%, and the patient factor $\hat{\beta}_{\bar{\theta}} = 0.97$ implies about 3%. This gap in annual discount rates is what generates the differential exit behavior between types documented throughout the calibration.

Both rates lie below the short-horizon experimental means for representative U.S. farmers. Duquette et al. (2012) estimate 28–43% across early and late adopters of conservation practices, with a pooled mean of 34%; the meta-review of Wuepper et al. (2023) reports an average of about 23%. Wuepper et al. (2023) also flag that short-horizon elicitation systematically overstate impatience relative to the multi-year horizons relevant for agricultural investments: when Grijalva et al. (2018) extend the convex time budget approach to a 20-year horizon, annual rates fall to 2–6%.

Our 3% patient discount rate sits inside this long-horizon range. It is close to the Bank of Canada’s long-run real neutral rate (Mendes 2014) and lies in the lower tail of the individual-level distribution documented for Denmark by Harrison et al. (2002). Our 11% impatient rate sits between the short- and long-horizon anchors. The gap between our patient rate and short-horizon experimental rates reflects two compounding distinctions: the elicitation-horizon adjustment emphasised by Wuepper et al. (2023), and the Money-Earlier-or-Later (MEL) versus utility-rate distinction of Cohen et al. (2020), where MEL experiments recover a required return on financial flows while structural estimates of the discount factor track a long-run utility rate that is mechanically lower.

²¹Relaxing the upper bound on $\hat{\beta}_\theta$ would push the estimate into a region with no empirical support. The sensitivity analysis in Appendix D.2 confirms that the qualitative results are stable across perturbations of $\hat{\beta}_\theta$ within a plausible range.

Both discount rates lying below the short-horizon farmer mean is consistent with positive patience selection at the organic-certification margin. The initial impatient share ($\hat{S}_0^\theta = 0.22$) should be read accordingly: it is the impatient mass remaining within a population already screened by the certification decision, which includes the three-year mandatory transition period, not a sample-wide estimate. The producers in our sample have already revealed enough patience to complete that three-year transition, so even our impatient type is more patient than the typical farmer in the experimental literature. That 22% of these pre-screened producers nevertheless remain present-biased underscores the strength of preference heterogeneity within the certified organic sector.

Pest-Shock States and Transition Dynamics. The moderate pest state implies a 10% yield reduction ($\hat{\sigma}_2 = 0.90$), within the 10–40% range documented for organic cereals (de Ponti et al. 2012, Ponisio et al. 2015). The severe state implies a 45% reduction ($\hat{\sigma}_3 = 0.55$), at the upper end of the range documented in the same meta-analyses for adverse pest realisations.

The calibrated transition matrix (Table 3) implies an expected sojourn of $1/(1 - \hat{\Omega}_{33}) \approx 6.7$ years in the worst pest regime ($\hat{\Omega}_{33} = 0.85$), consistent with the multi-year time scale on which soil-borne pathogen inoculum and disease-suppressive microbial communities evolve under organic management (Larkin 2015). Recovery from severe to moderate conditions ($\hat{\Omega}_{32} = 0.15$) and from moderate to good ($\hat{\Omega}_{21} = 0.20$) are both identified strictly away from their lower bounds, confirming that the data rather than the constraints drive recovery dynamics.

The persistence of the severe state is quantitatively important for the patience mechanism. With $\hat{\Omega}_{33} = 0.85$, adverse conditions persist long enough to drive out impatient producers yet remain recoverable enough that patient producers with $\hat{\beta}_\theta = 0.97$ assign substantial present value to the eventual return to mild conditions. The stationary-mean organic flow profit ($\hat{\omega}^\top \hat{\pi}_o^* \approx \$951/\text{ha}$)²² exceeds the conventional flow ($\hat{\pi}_c^* \approx \$824/\text{ha}$) by roughly \$127/ha. Capitalising at the type-specific discount factor, the long-run lifetime premium ($(\hat{\omega}^\top \hat{\pi}_o^* - \hat{\pi}_c^*)/(1 - \hat{\beta}_\theta)$) is about \$1,270/ha for impatient producers and \$4,233/ha for patients, a ~15% premium over the conventional outside option for both types.

Both types therefore expect organic to dominate conventional in the long run. What differs is the state-conditional comparison: in the severe pest state, the organic flow $\hat{\pi}_o^*(\sigma_3) = \$645/\text{ha}$ lies \$179/ha below the conventional flow. Impatient producers discount the eventual recovery too heavily for the deferred long-run premium to outweigh this immediate shortfall, and they exit; patient producers face the same shortfall but discount the future less, so the long-run premium dominates and they stay.

Logit Sensitivity. The estimated logit sensitivity $\hat{\gamma} = 0.002 (\$/\text{ha})^{-1}$ places exit decisions in the responsive middle of the admissible range. From Equation (3.12), the exit probability is $P_{t,\sigma_j}^\theta = 1/[1 + \exp(-\hat{\gamma} \Delta V)]$, where $\Delta V \equiv V_C^\theta - V_O^\theta(\sigma_j)$ is the dollar-per-hectare value gap favouring exit. A representative gap of $\Delta V = \$500/\text{ha}$ produces a logit argument $\hat{\gamma} \Delta V = 0.002 \times 500 = 1.0$, hence an exit probability of $1/(1 + e^{-1}) \approx 0.73$, or odds of exit equal to $e^1 \approx 2.7$. Near indifference ($\Delta V = 0$, so $P = 1/2$), the local slope of the exit probability with respect to the value gap is $dP/d\Delta V = \hat{\gamma} P(1 - P) = \hat{\gamma}/4 = 5 \times 10^{-4}$ per \$/ha; an additional \$100/ha of conventional advantage therefore raises the annual exit probability by approximately $5 \times 10^{-4} \times 100 = 5$ percentage points. Enterprises respond meaningfully

²² $\hat{\omega}$ denotes the long-run (stationary) distribution of the calibrated transition matrix $\hat{\Omega}$, the unique probability vector satisfying $\hat{\omega}^\top \hat{\Omega} = \hat{\omega}^\top$. Appendix C reports its calibrated values and its role in verifying condition (ii) of Proposition 1, namely that the long-run mean organic profit $\hat{\omega}^\top \hat{\pi}_o^*$ exceeds the conventional flow $\hat{\pi}_c^*$.

to profit differentials without collapsing to the deterministic threshold rule that $\hat{\gamma} \rightarrow \infty$ would imply, consistent with the model’s joint emphasis on financial incentives and unobserved heterogeneity.

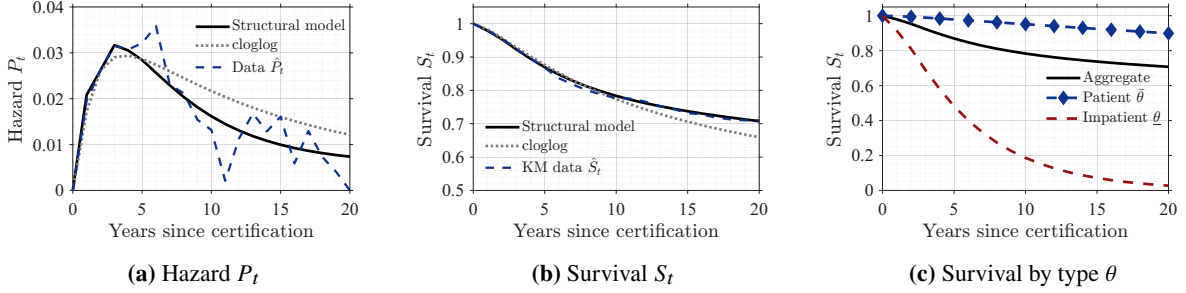


Figure 3: Model Fit: Data, cloglog, and Structural Model

Notes: This figure compares the structural model fit to the data. In panels (a) and (b), the dashed dark-blue line shows the raw observed hazard $\hat{P}_t$ (panel a) and the Kaplan–Meier survival $\hat{S}_t$ (panel b); the dotted-gray line shows the cloglog profile estimated on the crop-plant subsample from Equation (2.2); and the solid black line shows the structural model evaluated at the calibrated parameters $\hat{\chi}$ (Table 3). Panel (c) decomposes aggregate survival into the patient ($\bar{\theta}$, dashed dark blue) and impatient ($\underline{\theta}$, dashed dark red) sub-populations, each normalised to 1 at entry, with aggregate survival shown in solid black. Crop-plant sub-sample, *Portail Bio Québec*, $n_0 = 1,683$.

Model Fit and Selection. The structural model successfully matches the hump-shaped hazard profile, the primary empirical fact motivating the theoretical framework, with a peak around year 3 followed by a sustained decline. Panel (a) of Figure 3 compares the model-implied conditional hazard $P_t(\hat{\chi})$ to the raw observed hazard $\hat{P}_t$ and the cloglog-smoothed profile; the cloglog is not a calibration target but a validation benchmark, and its close agreement with the structural model confirms that smoothing does not introduce systematic bias. Panel (b) shows the corresponding aggregate survival $S_t(\hat{\chi})$ against the Kaplan–Meier estimate and the cloglog-smoothed survival; the model and Kaplan–Meier track closely throughout, while the cloglog-smoothed series (estimated on the restricted crop-plant sub-sample, $n_0 = 1,683$) departs from both after year 10, where the log-polynomial extrapolation diverges from the raw data as the risk set thins. Panel (c) decomposes aggregate survival by enterprise type: the impatient share falls from 22.0% at entry to near zero by year 15, while approximately 90% of the initial patient cohort remains certified at year 20. This compositional shift is the micro-foundation of the declining aggregate hazard established in Proposition 1.

Model Sensitivity and Validation. The role of heterogeneous time preferences cannot be directly verified in the data: discount factors are calibrated to aggregate survival moments rather than identified from observed preferences. Three pieces of evidence support the patience interpretation. First, certification in maple syrup or livestock (both activities requiring multi-year capital commitment that only forward-looking producers undertake) identifies long-horizon producers ex ante. Appendix D.1 shows non-parametrically (Figure D.2) that the Kaplan–Meier survival curve for long-horizon producers lies strictly above the short-horizon curve throughout the observed lifecycle, with a log-rank test rejecting equality of the two distributions, and reports the corresponding cloglog regressions, where the long-horizon coefficient is negative and significant across all four specifications. Second, the structural restriction tests in Appendix D.2 (Table D.4) show that any homogeneous-patience model (all impatient, all patient, or population-mean) fails to reproduce the inverted-U hazard, regardless of the value of the

common discount factor. Within the class of models considered, heterogeneous patience is necessary to generate the hump-shaped hazard; homogeneous-patience versions fail to match this feature. Third, the sensitivity grid in Appendix D.2 confirms quantitative stability across $\pm 20\%$ perturbations of pest persistence ($\hat{\Omega}_{33}$), organic profit ($\hat{\pi}_o^*$), and the two discount factors ($\hat{\beta}_\theta, \hat{\beta}_{\bar{\theta}}$).

4.3 Policy Analysis

Building on the calibrated model, we evaluate three policy instruments: (i) a post-certification subsidy (two variants, one-year and ten-year), (ii) a pesticide tax, and (iii) intensified pest pressure.²³ For each scenario we report aggregate survival dynamics and decompose the effect by producer type ($\theta, \bar{\theta}$).

Subsidy-Based Interventions. We consider two post-certification support schemes: (i) a one-off subsidy of \$100/ha in the first year following organic certification, consistent with the *Prime-Vert* program; and (ii) a transition subsidy of \$100/ha paid annually over the first ten years (Figure 4). Subsidies deliver lasting retention only when sustained over the full transition period (panel a, Figure 4).

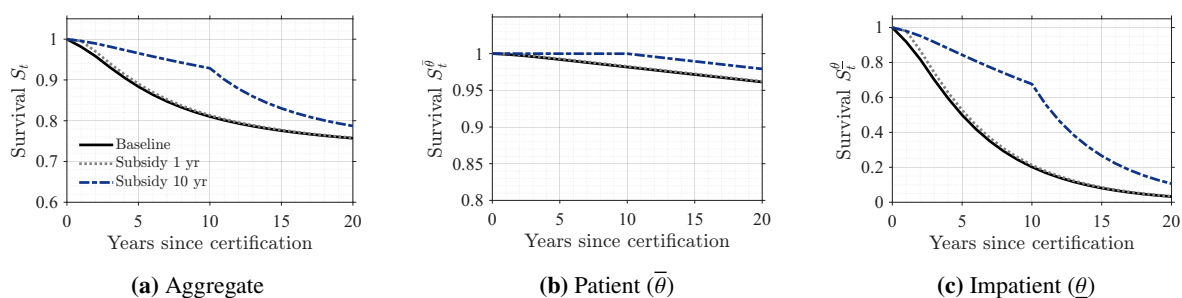


Figure 4: Survival Dynamics Under Subsidy Policies

Notes: This figure shows simulated survival S_t under three scenarios: the baseline (solid black), a one-year subsidy (dotted gray), and a ten-year subsidy (dash-dot dark blue). Panel (a) reports aggregate survival; panel (b) reports survival of patient enterprises ($\bar{\theta}$); panel (c) reports survival of impatient enterprises (θ). Each curve, baseline and both policy scenarios, is normalised to 1 at year 0 (entry). Subsidy amount: \$100/ha per year. Crop-plant sub-sample, $n_0 = 1,683$.

A one-year subsidy leaves ten-year survival essentially unchanged (81.3% versus 81.0% at baseline), because financial relief ends before the early-tenure peak in exits has run its course. A ten-year subsidy raises ten-year survival to 92.9%, but the gain does not persist: once the subsidy lapses, twenty-year survival reaches only 78.7% against 75.7% at baseline.

The gain is concentrated among impatient producers. Patient producers ($\bar{\theta}$) are near the ceiling at baseline and respond modestly: under the one-year subsidy, their ten-year survival rises from 98.2% to

²³The three scenarios differ deliberately in temporality, reflecting the nature of the instrument each represents rather than a data limitation. The one-year subsidy mirrors the actual design of Quebec's *Prime-Vert* programme, which pays organic-certification support as a one-time lump sum rather than a recurring annual transfer; the ten-year variant is a counterfactual extension, evaluating how the same per-hectare support would perform if sustained annually over a much longer commitment window than the programme actually offers. An indefinite subsidy is not modelled, since no comparable programme design maintains open-ended per-hectare transfers. The pesticide tax, by contrast, is a permanent change to the relative price of a conventional input rather than a bounded transfer, so it is modelled as a standing policy with no expiration date. The pest-pressure scenario is likewise permanent, representing a structural, climate-driven deterioration in yields rather than a transient shock. Because all three scenarios report survival at every year, the ten-year subsidy and the permanent tax remain directly comparable at any common horizon, even though only the subsidy's own duration is bounded.

98.3%; under the ten-year subsidy, it reaches nearly 100% through year ten and settles at 97.9% by year twenty (panel b, Figure 4). Impatient producers (θ) start from a baseline ten-year survival of 20.2%, and the ten-year subsidy raises this to 67.6%, a gain of 47 percentage points (panel c, Figure 4). Once the subsidy lapses, however, impatient survival regresses sharply toward the baseline trajectory, and twenty-year survival approaches the baseline level. The subsidy retains impatient producers only as long as it is paid.

Tax-Based Interventions. We simulate an 80% increase in pesticide cost to conventional producers ($\tau = 0.80$), consistent with the *Sustainable Agriculture Plan* (SAP) target of roughly a 12% reduction in pesticide use (Séguin & Thiam 2025). An 80% pesticide tax raises ten-year survival from 81.0% to

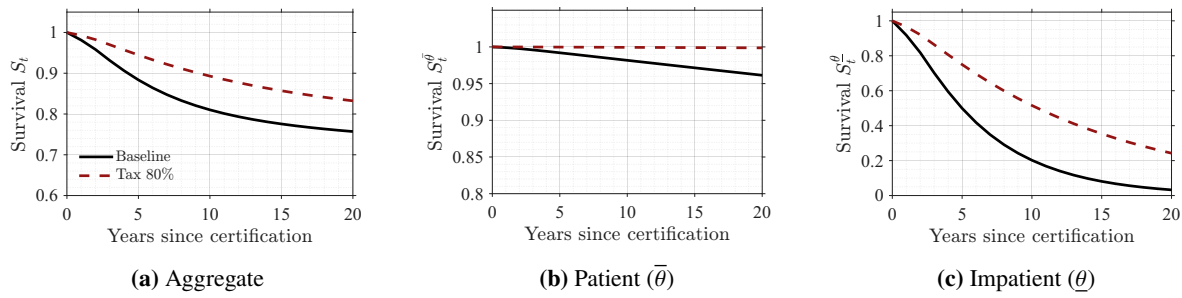


Figure 5: Survival Dynamics Under Pesticide Tax Policy

Notes: This figure shows simulated survival S_t under two scenarios: the baseline (solid black) and an 80% pesticide tax (dashed dark red). Panel (a) reports aggregate survival; panel (b) reports survival of patient enterprises; panel (c) reports survival of impatient enterprises. Each curve, baseline and policy scenario, is normalised to 1 at year 0 (entry). The tax rate is calibrated to achieve the 12% reduction in pesticide use targeted by the *Sustainable Agriculture Plan* (Séguin & Thiam 2025). Crop-plant sub-sample, $n_0 = 1,683$.

89.3% and twenty-year survival from 75.7% to 83.2% (panel a, Figure 5). Unlike the subsidy, the gain persists: the tax permanently raises the cost of the conventional alternative, so producers face the higher incentive to stay organic throughout the lifecycle.²⁴

The tax pulls both types toward full retention, but impatient producers move the most. Patient producers ($\bar{\theta}$) are near the ceiling at baseline and see a small upward shift: their ten-year survival rises from 98.2% to 99.9% and twenty-year survival from 96.1% to 99.9% (panel b, Figure 5); the tax eliminates the residual temptation to revert. Impatient producers (θ) start from a baseline ten-year survival of 20.2%, and the tax raises this to 51.6%, a gain of 31 percentage points (panel c, Figure 5). Because the tax does not lapse, impatient producers retain this gain across the full lifecycle rather than regressing toward baseline.

Biophysical Risks and Pest-Related Shocks. We simulate a 1% uniform reduction in all pest-state productivities, yielding new shock values $\{0.99, 0.89, 0.54\}$. The magnitude is deliberately small, but its economic footprint is not negligible: at CRAAQ prices, a one-percent yield cut translates to losses of roughly \$45/ha for corn, \$33/ha for soybean, and \$12/ha for barley.²⁵ With no aggregate mean-reversion,

²⁴The tax counterfactual is partial equilibrium: the organic pest-shock distribution Ω is treated as exogenous and p_x is fixed; x adjusts via Equation (3.8) as $(1 + \tau)p_x$ rises. Changes in regional pest pressure from altered pesticide use are not modelled. The results therefore capture the tax's direct incentive effect on the organic–conventional profit gap, ceteris paribus.

²⁵For the three organic field crops in Table D.1, the one-percent cut removes 0.091 t/ha from corn (priced at \$490/t), 0.029 t/ha from soybean (at \$1,151/t), and 0.027 t/ha from barley (at \$458/t). A larger magnitude would scale the response

the 1% loss is certain and permanent: the producer cannot recover it through better input choices or wait it out. The scenario captures system-wide biophysical deterioration—climate-driven pest incidence or resistant strains—rather than idiosyncratic variability. A 1% productivity loss lowers aggregate ten-year

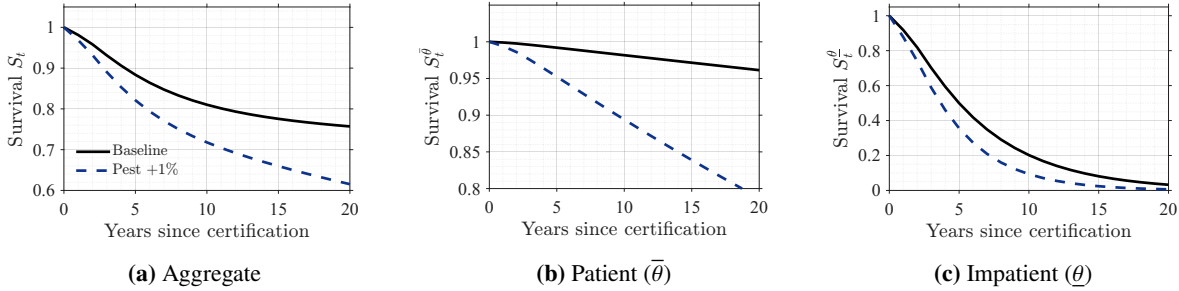


Figure 6: Survival Dynamics Under Intensified Pest Pressure

Notes: This figure shows simulated survival S_t under two scenarios: the baseline (solid black) and an intensified pest-pressure scenario (dashed dark blue). Panel (a) reports aggregate survival; panel (b) reports survival of patient enterprises; panel (c) reports survival of impatient enterprises. Each curve, baseline and policy scenario, is normalised to 1 at year 0 (entry). The pest-pressure scenario applies a 1% uniform reduction in all pest-state productivities $\{\sigma_i\}$, yielding $\{0.99, 0.89, 0.54\}$. Crop-plant sub-sample, $n_0 = 1,683$.

survival from 81.0% to 71.8% and twenty-year survival from 75.7% to 61.5% (panel a, Figure 6). The pest scenario shifts the survival curve down structurally rather than displacing it temporarily: because the productivity loss affects every state of nature, no recovery emerges over the twenty-year horizon.

Unlike the subsidy and the tax, the pest shock hits both types. Patient producers ($\bar{\theta}$) lose their forward-looking advantage: their ten-year survival falls from 98.2% to 89.4% and twenty-year survival from 96.1% to 78.7% (panel b, Figure 6). The permanent productivity loss pulls the long-run value of organic production below the conventional alternative even for patient producers, and exit becomes the optimal response. Impatient producers ($\underline{\theta}$) collapse outright: their ten-year survival falls from 20.2% to 9.3% and stays there (panel c, Figure 6). The pest shock compresses their already-short payoff horizon, eliminating the window over which staying organic remains worthwhile. Pest-risk mitigation instruments are therefore not substitutes for financial support but necessary complements, particularly for preserving the patient segment whose long-run retention underpins sector stability.

Although discount factors are not directly observable in administrative data, policy can target an observable proxy for patience: enterprises certified in long-horizon production categories such as maple/forest or livestock self-select on patience and survive substantially longer than the rest of the sector (Appendix D.1).

5 Conclusion

This paper examines the dynamics of exit from organic certification in Quebec, the mechanisms underlying its lifecycle pattern, and their implications for retention policy. Using enterprise-level certification records from the *Portail Bio Québec*, we combine survival econometrics with a dynamic model of the organic–conventional production choice. The empirical analysis shows that exit is concentrated in the first five years and follows a hump-shaped hazard, peaking around the third year before declining. Exit

mechanically and make the structural sensitivity harder to read; by picking 1% we preserve all structural parameters and isolate the dynamic amplification effect of the permanent productivity loss.

risk also varies systematically across enterprises: diversification across certified product categories is associated with lower exit risk, while maple/forest and livestock certifications are associated with greater retention and wild-harvest certification with higher exit risk. These patterns are stable across exit cohorts and indicate that organic retention depends on both the stage of the certification lifecycle and enterprise characteristics.

The dynamic model provides a structural interpretation based on heterogeneous intertemporal preferences and persistent pest shocks. Producers with shorter planning horizons respond more strongly to adverse productivity shocks and are more likely to exit, whereas patient producers place greater weight on the longer-run returns to organic production and remain certified. As impatient producers exit, the composition of survivors shifts toward patient producers, generating the declining portion of the aggregate hazard. The calibrated model implies that approximately 22% of entrants belong to the impatient type, with an implied annual discount rate of about 11%, compared with roughly 3% for the remaining 78% of entrants, who are more patient.

The policy simulations suggest that these mechanisms matter for retention policy. A one-year post-certification subsidy has little effect on long-run survival, with ten-year survival increasing from 81.0% to 81.3%, whereas extending the subsidy for ten years raises survival to 92.9%. The effect of temporary support therefore largely dissipates after the subsidy ends. A pesticide tax produces a more persistent effect by permanently changing the relative attractiveness of the conventional alternative: an 80% tax raises ten-year survival to 89.3%. Neither instrument, however, eliminates biophysical risk. A permanent 1% productivity loss from intensified pest pressure reduces ten-year survival to 71.8%, indicating that financial support and pest-risk mitigation may operate as complements rather than substitutes.

Several caveats remain. First, the enterprise-level panel covers only three years, so longer-run dynamics necessarily rely partly on model extrapolation. Second, several structural parameters are calibrated rather than directly estimated from micro-level measures of preferences or pest exposure; the model should therefore be interpreted as a disciplined mechanism for understanding the observed hazard rather than as a direct measurement of producer preferences. Finally, the analysis focuses on exit conditional on certification and abstracts from entry and general-equilibrium effects. Future research could extend the framework by jointly modelling entry and exit, incorporating general-equilibrium feedbacks from pesticide taxation, and linking certification records to enterprise-level measures of productivity, financial conditions, and spatially correlated pest shocks.

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Online Appendix

Exit Dynamics from Organic Certification in Quebec

Komla Avoumatsodo Moustapha Thiam

A Descriptive Appendix

This appendix reports the international and Canadian context of Quebec’s organic sector, the structural composition of the sector by product category, certifier and administrative region, and the disaggregated net-change and entry–exit distributions that support the common-shock reading of the 2025 contraction in Section 2.1. The distributions shown below are quantitatively stable across 2023–2025 and therefore serve as fixed descriptive inputs rather than moving parts of the main analysis; the disaggregated 2025 flows, by contrast, are what rule out a regional, compositional, or size-based reallocation story for the contraction.

A.1 International and Canadian Context

Quebec’s organic sector operates within an uneven international landscape. Data from the Research Institute of Organic Agriculture (FiBL) yearbook (Willer et al. 2024) show substantial cross-country disparities in land devoted to organic practices. Australia leads with roughly 53 million hectares under organic management, followed by India (4.4 million ha), Argentina (4 million ha) and Uruguay (3.5 million ha), while Canada has a more modest footprint of about 1.3 million hectares. Share-based comparisons give a different ranking: Liechtenstein (44.5%), Austria (27%), Uruguay (25%) and Estonia (22%) top the list in terms of the share of agricultural land under organic cultivation, whereas Canada sits at about 2%, below the European Union average of roughly 4% (Figure A.1).

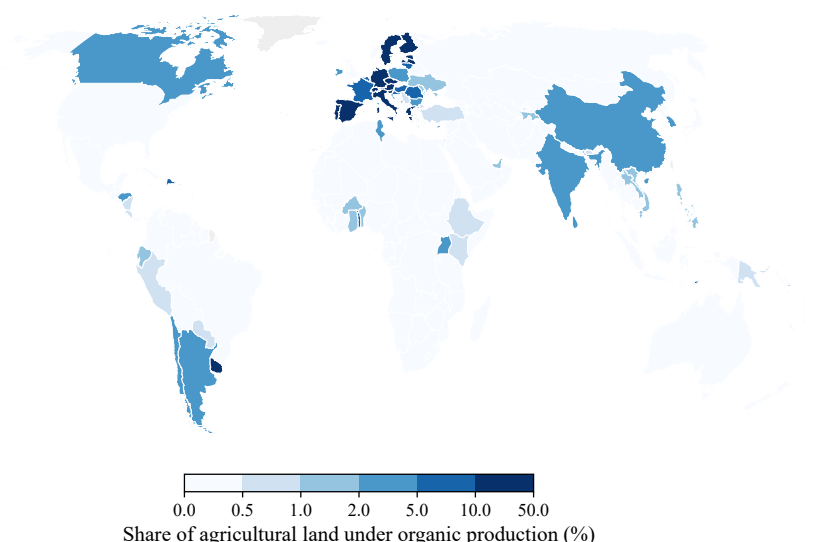


Figure A.1: Percentage of Total Agricultural Land Devoted to Organic Farming (2023)

Notes: This figure shows the percentage of agricultural land in each country dedicated to organic production in 2023. *Source:* FiBL (Willer et al. 2024).

Within Canada, Quebec occupies a distinctive position. With 8.4% of its agricultural enterprises certified organic, Quebec is the national leader and substantially exceeds the rates observed in Manitoba, Saskatchewan, Alberta and Nova Scotia (Figure A.2). Quebec alone accounts for approximately 82% of Canada’s organic fruit production area and 65% of the organic vegetable area.

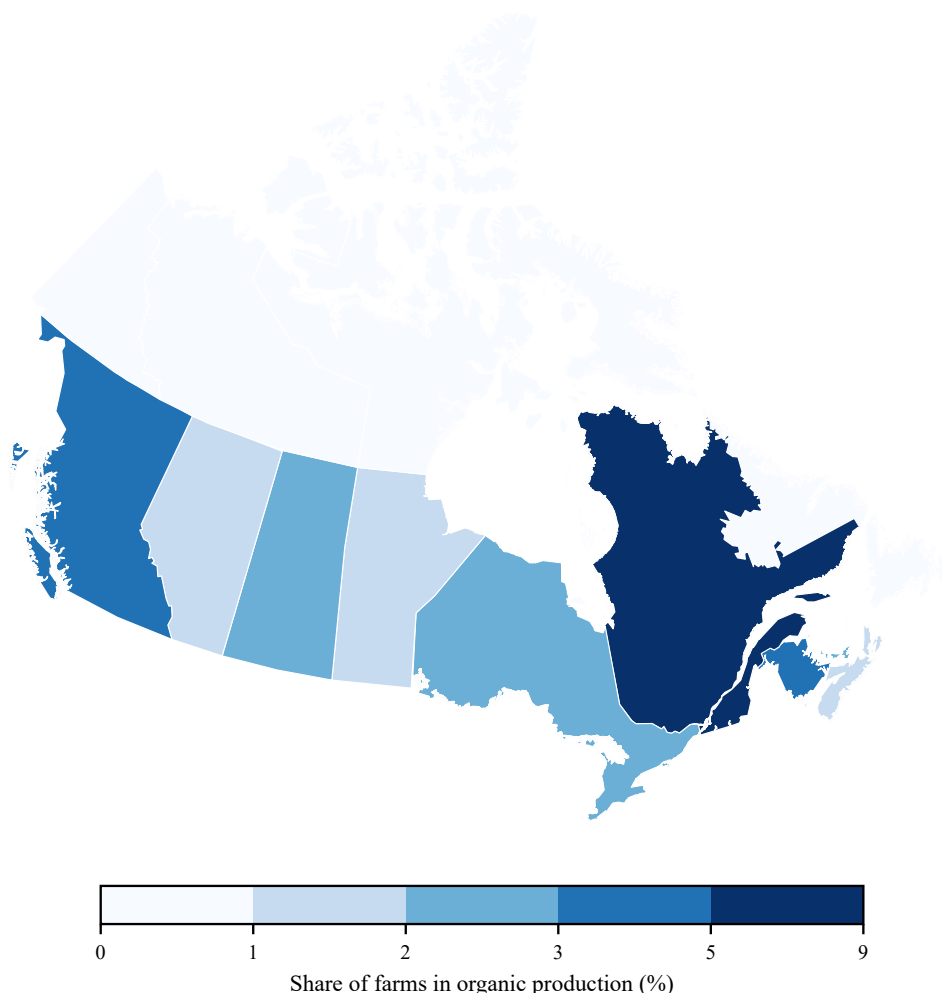


Figure A.2: Percentage of Farms in Organic Production by Canadian Province (2021)

Notes: This figure shows the percentage of agricultural farms engaged in organic production in each Canadian province in 2021. Quebec leads at 8.4%, while other provinces cluster between 1% and 4%. *Source:* Statistics Canada.

A.2 Structural Composition of the Quebec Organic Sector

The three distributions across product categories, certifying bodies and administrative regions are stable over 2023–2025. Figure A.3 reports that crop-plant production accounts for roughly half of all certifications in every year, followed by maple (27–30%), processed products (13%) and livestock (5.6%). Figure A.4 shows that Ecocert Canada holds about 86% of the certification market in each year, with Québec-Vrai retaining 11–12%; the remaining four certifiers together account for less than 2%. Figure A.5 reports the regional distribution: Chaudière-Appalaches, Estrie, Montérégie, Bas-Saint-Laurent and Centre-du-Québec account for more than two thirds of certifications in every year. Because none

of these distributions move materially between 2023 and 2025, the 2025 contraction documented in Figure 1 cannot be explained by a shift in the certifier, product, or regional composition.

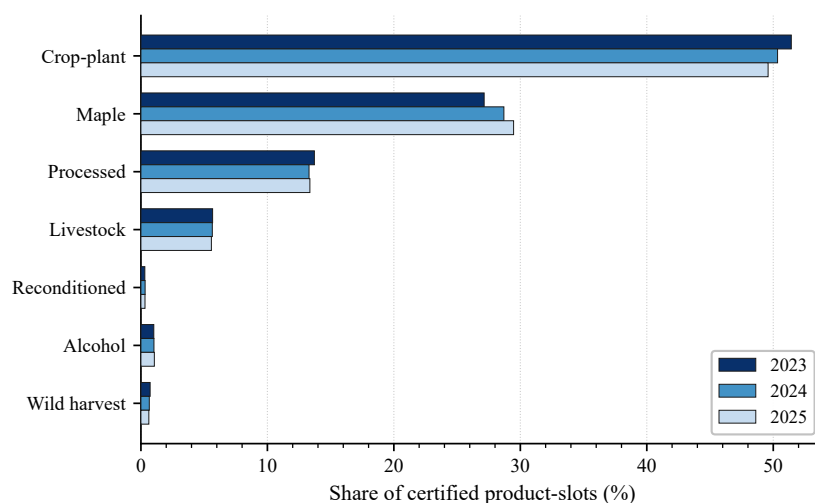


Figure A.3: Certified Enterprises as Percentage of Total by Category in Quebec, 2023–2025

Notes: This figure shows the distribution of certified organic enterprises by product category in Quebec, expressed as a percentage of the total for 2023, 2024 and 2025. Categories: alcoholic beverages, animal production, food preparation, plant-based crops, reconditioned products, wild harvest, and maple products.

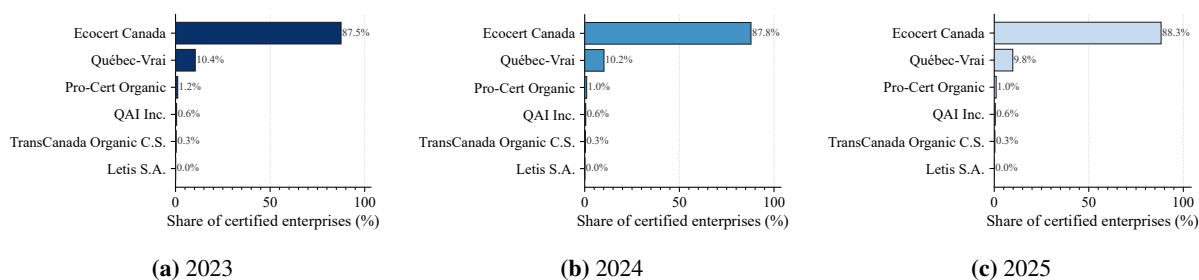


Figure A.4: Percentage of Certified Enterprises by Certifier in Quebec

Notes: This figure shows the share of total organic certifications attributed to each certifying body in Quebec for 2023, 2024 and 2025.

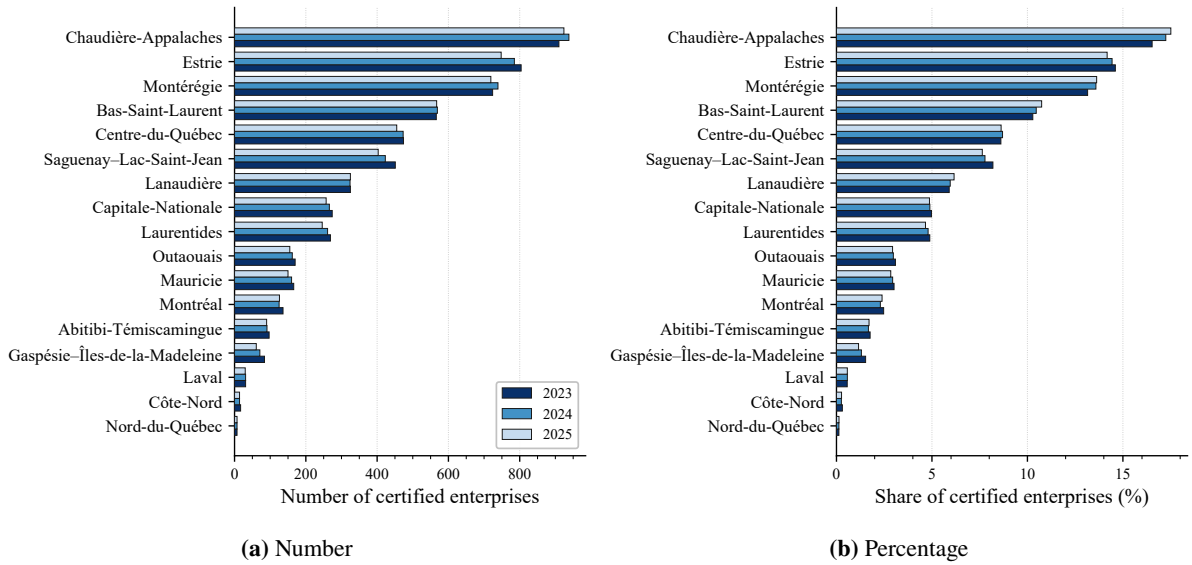


Figure A.5: Organic Enterprises Certified by Region in Quebec, 2023–2025

Notes: This figure shows the regional distribution of organic certifications in Quebec for 2023, 2024 and 2025. Panel (a) reports the number of certifications held by organic enterprises across Quebec’s administrative regions; panel (b) reports the corresponding within-year percentage shares. The legend on panel (a) applies to both panels.

A.3 Disaggregated Net Change and Entry–Exit Flows

The aggregate 2025 contraction documented in Figure 1 is spatially and compositionally uniform rather than concentrated in a subset of products or regions. Figure A.6 shows the net change in certified enterprises by product category: in 2023 the maple category grew by 11.3% of base, offsetting net losses of 1.7% in crop-plant and 2.9% in processed products; in 2024 the maple buffer thinned to 4.3%; in 2025 every major product category except reconditioning registers a net outflow, and maple itself flips to -0.4% . Figure A.7 documents the same pattern at the entrant level: in 2023 single-product entrants outnumbered single-product exiters by 245 to 152 (a 1.6-to-1 ratio); in 2024 the ratio is $132/148 \approx 0.9$; in 2025 it falls to $83/132 \approx 0.6$. Because single-product certifications are the entry gateway, their decline is a leading indicator of aggregate inflow weakness. Figure A.8 provides the spatial counterpart: exits exceed entries in 6 of Quebec’s 17 administrative regions in 2023, 12 of 17 in 2024, and 13 of 17 in 2025, with the deterioration strongest in the historic centres (Chaudière-Appalaches moves from net +38 to -14 , Estrie from +30 to -37 , Montérégie from +8 to -20).

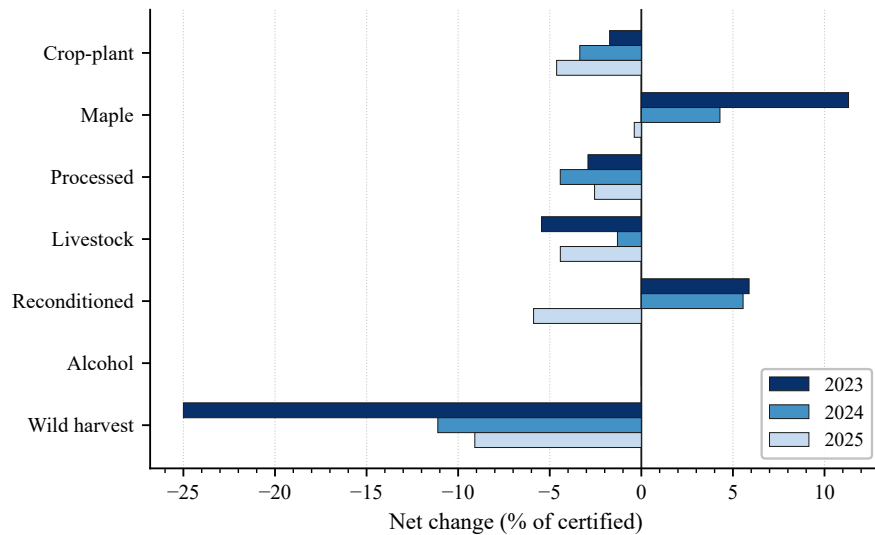


Figure A.6: Net Change as Percentage of Certified Enterprises by Category

Notes: This figure shows the net change in certifications, expressed as a percentage of the previous year's certification stock, for each product category in 2023, 2024 and 2025.

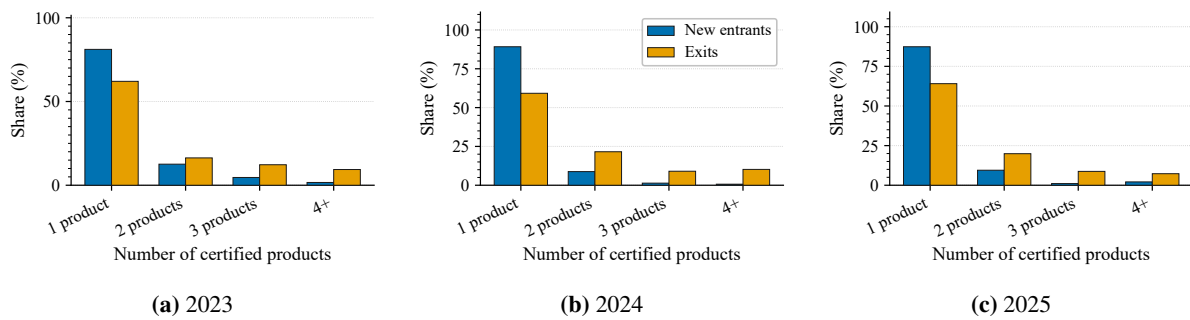


Figure A.7: Entries and Exits by Number of Products

Notes: This figure shows the distribution of enterprises by the total number of certified products held at entry into or exit from the organic sector, for 2023, 2024 and 2025 (counts summed across the seven product categories). Blue bars: share of new entrants holding n certified products (as a percentage of total entrants); amber bars: share of exiting enterprises by the same classification. The legend on the 2024 panel applies to all three panels.

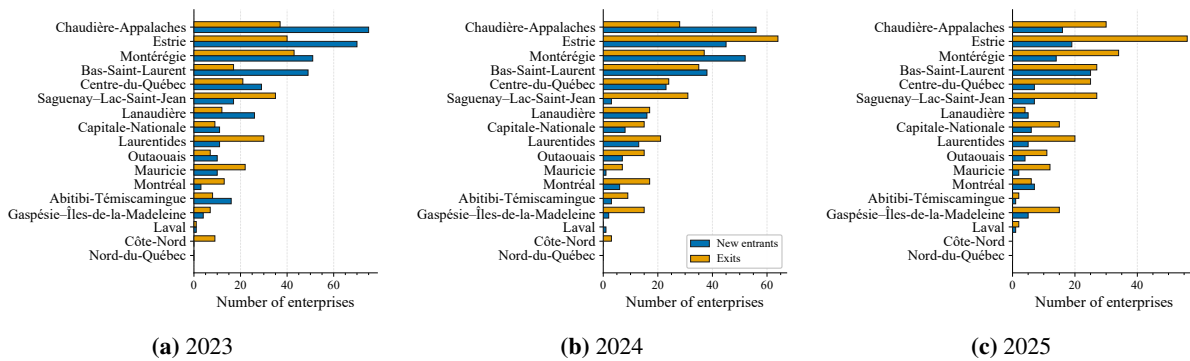


Figure A.8: Entries and Exits by Region

Notes: This figure shows new entrants (blue) and exits (amber) by Quebec administrative region for 2023, 2024 and 2025. Regions are ordered by total certified count in 2024. The legend on the 2024 panel applies to all three.

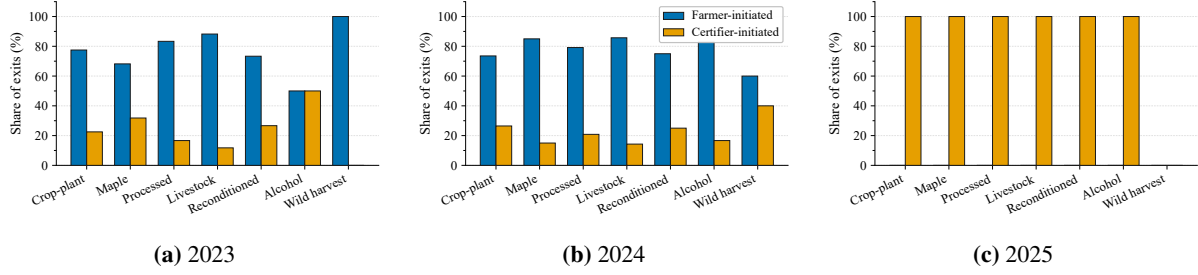


Figure A.9: Distribution of Exits by Product and Initiator

Notes: This figure shows the share of exits for each product, broken down by initiator (producer vs. certifier), expressed as a percentage of exits whose initiator is known. The legend on the 2024 panel applies to all three panels. The 2025 panel reports the 34 of 206 exit records for which an initiator was recorded in the *Portail Bio Québec* registry at the time of extraction; for every product category with a known initiator the exit was classified as certifier-initiated, so the 2025 panel should be read as a lower bound on the certifier share rather than as a structural shift.

A.4 Kaplan–Meier Survival and Cloglog Smoothed Hazard

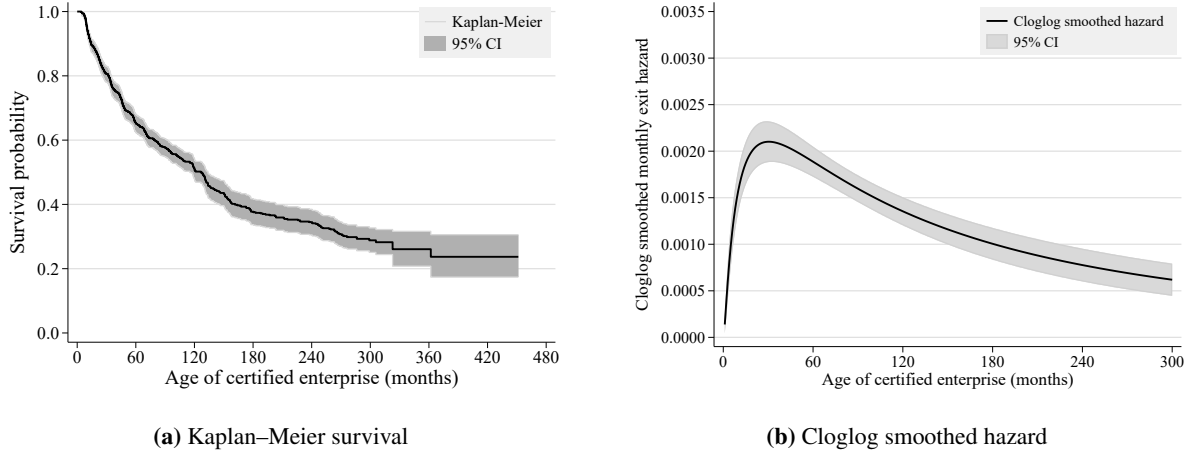


Figure A.10: Survival and smoothed hazard for Quebec organic enterprises

Notes: This figure shows the survival function and the smoothed monthly exit hazard estimated on the whole sample of certified enterprises. Panel (a) reports the Kaplan–Meier estimate of the survival function (proportion of enterprises still certified), using all enterprise-entry spells observed 2023–2025 with left-truncation adjustment for spells that began before 2023. Panel (b) reports the cloglog smoothed monthly exit hazard (solid line), $\hat{h}(t) = 1 - \exp(-\exp(\hat{\beta}_0 + \hat{\beta}_1 \ln t + \hat{\beta}_2 \ln^2 t))$, with 95% pointwise confidence band (shaded) obtained by the delta method from the cluster-robust covariance; estimated on the enterprise-month panel that underlies the cloglog estimation in Table 1. The same parametric profile (without confidence band) is plotted as the dotted-gray line in Figure 3.

Figure A.10 reports two objects computed from the enterprise-month panel underlying the cloglog estimation. Panel (a) is the Kaplan–Meier survival function $\hat{S}_t = \prod_{s=1}^t (1 - \hat{P}_{s-1})$, the cumulative probability that an enterprise remains certified through duration t ; it falls steeply during the early years and flattens from roughly the ten-year mark onward, mirroring the cumulative exit distribution of Figure 2 (panel a) which also reaches the bulk of the realised mass by that horizon. Panel (b) is the cloglog smoothed hazard, the parametric log-polynomial fit $\hat{h}(t) = 1 - \exp(-\exp(\hat{\beta}_0 + \hat{\beta}_1 \ln t + \hat{\beta}_2 \ln^2 t))$, displayed with a 95% pointwise confidence band obtained by the delta method from the cluster-robust covariance. The smoothed hazard reproduces the inverted-U shape estimated in Section 2.3, peaking near three years and declining thereafter; the confidence band widens with age as the risk set thins.

Figure A.11 disaggregates the pooled 2023–2025 age-at-exit distribution by certified product category. The density is computed within each category-specific sample of exits and superposed on the pooled total (solid grey). Across crop-plant, maple/forest, processed, livestock, and alcohol, the early-tenure peak is preserved: between 46% and 67% of exits occur within sixty months of certification, mirroring the inverted-U documented in Figure 2. The reconditioned-products category is dropped because only two exits are observed in 2023–2025.

Wild harvest is the one exception. While the other five densities turn down after the 43-month peak identified in Section 2.3, the wild-harvest density continues to rise past that point and reaches its peak only around seven years. Exit risk for wild-harvest enterprises therefore keeps increasing over the range where every other category is already declining. This is the unconditional counterpart of the positive wild-harvest coefficient in Table 1: the elevated exit hazard the table identifies conditional on age is precisely the mechanism that prevents the density from turning down. Setting wild harvest aside, the robustness of the inverted-U across all remaining categories rules out an interpretation in which the lifecycle shape is driven by a single dominant category.

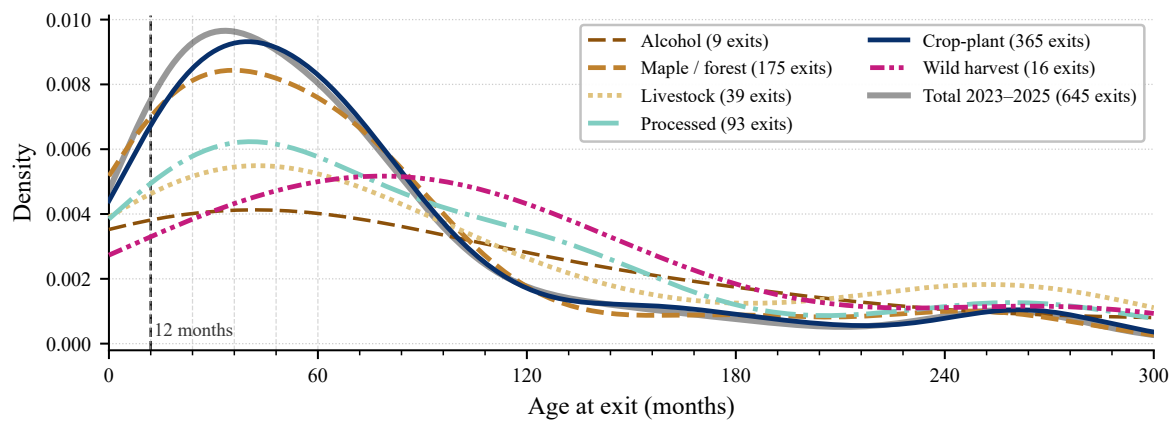


Figure A.11: Density of age at exit, by certified product category (pooled 2023–2025)

Notes: This figure shows kernel-density estimates of the age at exit for enterprises holding at least one certification in each of the seven product categories, computed on the pooled 2023–2025 exit sample. Each enterprise contributes to every category in which it holds a certification, so the curves are not mutually exclusive. The thick semi-transparent grey curve is the pooled distribution across all 668 observed exits. The number of exits underlying each category-specific density is reported in the legend. The reconditioned-products category is omitted because only two exits are observed in the sample, too few for a kernel density estimate. The vertical dashed reference line highlights the 12-month anniversary of certification.

B Econometric Appendix

This appendix complements Section 2 in three ways. Section B.1 documents the data constructions underlying the three estimators in Table 1 and reports the logit discrete-time hazard equation explicitly. Section B.2 disaggregates the aggregate diversification gradient (Table 2, column 1) by product subsample. The remaining sections probe the main specification along five complementary dimensions.

Section B.3 decomposes the diversification margin into a breadth component, which counts distinct certified product categories, and a depth component, which counts extra products within each already-certified category, and enters the two jointly. Section B.5 replaces the binary product indicators of

Table 1 with raw per-category counts, and Section B.6 groups those counts into zero-, one-, and two-or-more bins per category. Section B.7 restricts the sample to enterprises whose first certification falls inside the observation window, eliminating potential left-truncation bias; and Section B.8 replaces the log-polynomial duration controls with non-parametric age-band indicators. In every exercise the sign, magnitude, and significance of the product-type and diversification effects remain essentially unchanged.

B.1 Survival Data Construction and Discrete-Time Hazard Models

The Cox, cloglog, and logit specifications in Table 1 are estimated on two different data constructions, which explain the differing observation counts.

Enterprise-spell format (Cox). Each enterprise contributes a single row recording its observed duration T_i in months, an event indicator (1 = exit observed, 0 = right-censored), and the time-invariant covariates X_i (17 administrative regions, 4 entry-cohort bins, 6 certifiers, and 7 product categories). The Cox partial likelihood handles left truncation by restricting the risk set at each event time t to enterprises whose entry into the observation window is at most t and whose duration is at least t . Estimation uses $n = 4,100$ enterprises.

Table B.1: Enterprise-spell data layout (Cox)

Enterprise	Duration	Event	Region	Cohort
1	4	1	Estrie	2023
2	60	1	Montréal	2018
3	120	0	Bas-Saint-Laurent	2014
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
4,100	12	1	Chaudière-Appalaches	2024

Notes: The table illustrates the enterprise-spell layout used by the Cox specification, with one row per enterprise. *Duration* is the observed spell length in months and *Event* equals 1 if exit is observed and 0 if the spell is right-censored at the end of the observation window. *Region* and *Cohort* stand in for the full vector of time-invariant covariates X_i (region, entry cohort, certifier, product category); *Cohort* denotes the calendar year in which the enterprise was first certified organic. The Cox partial likelihood handles left truncation by restricting the risk set at each event time to enterprises with entry at most t and duration at least t .

Enterprise-month format (cloglog and logit). Each enterprise’s spell is reconstructed from first certification (month $t = 1$) through its observed exit or end-of-window censoring, with one binary observation per month: $y_{it} = 1$ in the month of exit and $y_{it} = 0$ in every other month. For cohorts certified before the 2023 observation window, the pre-window months enter the panel as $y_{it} = 0$ (survival) observations: appearance in the 2023 registry under a recorded certification date implies continuous certification from that date through 2023, so those months are filled by registry-implied survival rather than directly observed. Entry-cohort fixed effects absorb any residual heterogeneity across entry years; the panel does not condition on window entry in the partial-likelihood sense that the Cox specification does. Using the same enterprises as above (enterprise 1 exits at month 4, enterprise 2 at month 60, enterprise 3 is censored at month 120, enterprise 4 exits at month 8), the panel takes the following form. The same information appears in two formats: enterprise 1 generates 4 rows in the cloglog/logit panel

Table B.2: Enterprise-month panel layout (cloglog and logit)

Enterprise	Month	Exit	Age	Region	Cohort
1	1	0	1	Estrie	2023
1	2	0	2	Estrie	2023
1	3	0	3	Estrie	2023
1	4	1	4	Estrie	2023
2	1	0	1	Montréal	2018
2	2	0	2	Montréal	2018
⋮	⋮	⋮	⋮	⋮	⋮
2	60	1	60	Montréal	2018
3	1	0	1	Bas-Saint-Laurent	2014
3	2	0	2	Bas-Saint-Laurent	2014
⋮	⋮	⋮	⋮	⋮	⋮
3	120	0	120	Bas-Saint-Laurent	2014
4	1	0	1	Lanaudière	2025
⋮	⋮	⋮	⋮	⋮	⋮
4	8	1	8	Lanaudière	2025

Notes: The table illustrates the enterprise-month panel layout used by the cloglog and logit specifications, with one row per enterprise-month at risk. *Exit* equals 1 in the month of exit and 0 otherwise, and *Age* is the duration since first certification. *Region* and *Cohort* stand in for the full vector of time-invariant covariates X_i ; *Cohort* denotes the calendar year in which the enterprise was first certified organic. Right-censored enterprises contribute zeros throughout (enterprise 3); enterprises that exit contribute a single one in the month of exit (enterprises 1, 2, 4). Recent entry cohorts (2023, 2024, 2025) generate short panels because total tenure since first certification is short.

(months 1 to 4) but a single row in the Cox file (duration = 4, event = 1). This expansion converts the discrete-time hazard problem into a binary regression on y_{it} , and applying the cloglog or logit link recovers the per-period hazard $h_{it} = \Pr(T_i = t \mid T_i \geq t)$. The cohort-FE specification yields $n = 423,362$ enterprise-month rows; the calendar-year-FE specification yields $n = 154,718$, the reduction arising because enterprise-months in calendar years where all enterprises either exit or survive are dropped to avoid perfect prediction (standard in nonlinear panel models with fixed effects).

Logit discrete-time hazard. The logit specification reported in Table 1 replaces the cloglog link in Equation (2.2) with the logit link, holding the right-hand side fixed:

$$\log\left(\frac{h_{it}}{1 - h_{it}}\right) = \beta_1 \ln(t) + \beta_2 \ln^2(t) + X_i' \alpha. \quad (\text{B.1})$$

For the small per-month exit hazards observed in our data (h_{it} on the order of 10^{-3}), the logit and cloglog links produce numerically near-identical coefficients, since $-\log(1 - h) \approx h \approx h/(1 - h)$ as $h \rightarrow 0$.

Treatment of right-censoring and left-truncation. The two data constructions handle right-censoring and left-truncation differently in form but identically in econometric content.

The Cox estimator absorbs censoring through the risk set. The partial likelihood

$$\mathcal{L}_{\text{Cox}} = \prod_{i: \text{event}_i=1} \frac{\exp(X'_i \alpha)}{\sum_{j \in \mathcal{R}(t_i)} \exp(X'_j \alpha)}, \quad (\text{B.2})$$

where the outer product runs over enterprises i that exit (event = 1) at their observed time t_i , and the denominator sums over the risk set $\mathcal{R}(t_i) = \{j : \text{enterprise } j \text{ is at risk at time } t_i\}$, that is, all enterprises that have neither exited nor been censored before t_i and that have already entered the observation window by t_i . A right-censored enterprise enters this denominator at every event time it survives but never the numerator: it contributes the information that it has not yet exited, with no terminal event. Pre-2023 cohorts are handled through delayed entry: an enterprise first certified before the 2023 window opens enters $\mathcal{R}(\cdot)$ only from its window-entry duration onward, so the partial likelihood conditions on survival to that point and the early-duration risk sets are populated only by enterprises certified within the observation window.

The cloglog estimator absorbs right-censoring through the panel construction itself. Each enterprise contributes one row per month at risk, with $y_{it} = 0$ in every survival month and $y_{it} = 1$ in the month of exit; right-censored enterprises simply stop generating rows after their last observed month, with no terminal $y_{it} = 1$ row (enterprise 3 in Table B.2). The discrete-time likelihood

$$\mathcal{L}_{\text{cloglog}} = \prod_{i,t} h_{it}^{y_{it}} (1 - h_{it})^{1-y_{it}} \quad (\text{B.3})$$

runs over enterprise i and duration t since first certification, and treats every survival row $y_{it} = 0$ as a non-event regardless of whether it falls within the 2023–2025 observation window. Pre-2023 cohorts are handled differently than in the Cox specification: the cloglog panel reconstructs each spell from first certification ($t = 1$) onward, and pre-window months for these older cohorts enter as $y_{it} = 0$ rows. The justification is the registry-implied continuous-certification assumption described above: appearance in the 2023 registry under a recorded entry date implies that the enterprise was continuously certified, and therefore did not exit, in every month between that entry date and 2023. Entry-cohort fixed effects absorb residual between-cohort heterogeneity in pre-window survival.

Both estimators rest on the same identifying assumption that censoring is non-informative, that is, the closing year 2025 is independent of the residual exit hazard conditional on X_i . Under this assumption the Cox and cloglog estimators are asymptotically equivalent, which is consistent with the close agreement between the two columns in Table 1.

B.2 Diversification Effect by Product Subsample

Figure B.1 illustrates the aggregate diversification gradient that underlies the column-(1) estimate of Table 2. The predicted monthly exit probability declines monotonically as an enterprise accumulates additional distinct certified product categories, giving a non-linear counterpart to the single average marginal effect reported in the main text.

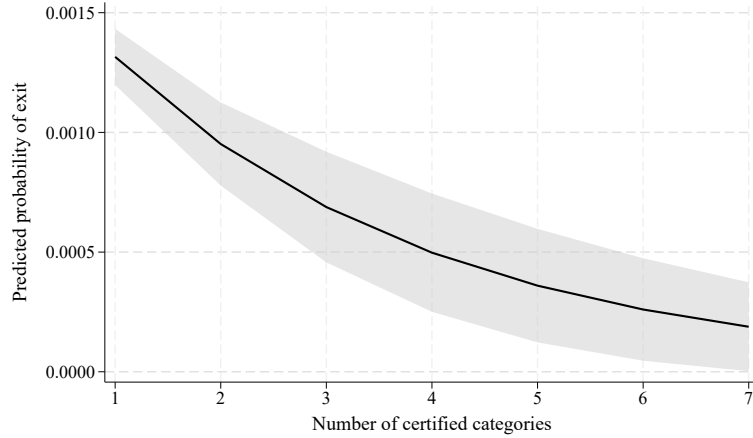


Figure B.1: Predicted exit probability by number of certified product categories (aggregate)

Notes: This figure shows the predicted monthly exit probability from the pooled cloglog model with entry-cohort fixed effects, plotted against the number of distinct certified product categories; the shaded band is the 95% confidence interval, with all other covariates held at observed values. The underlying regression is the same as in column (1) of Table 2. All specifications include region, certifier, and entry-cohort controls; standard errors are clustered at the enterprise level.

Table B.3 re-estimates the diversification specification of column (1) of Table 2 separately on each product-type subsample. Each column restricts the enterprise-month panel to enterprises holding at least one certification in the indicated category and reports the cloglog average marginal effect of the number of distinct certified product categories, directly comparable in units to the aggregate estimate of -0.0005 in column (1) of Table 2. Standard errors are clustered at the enterprise level, and region, certifier, and entry-cohort fixed effects enter as in the main specification. The maple/forest and organic basket subsamples are omitted because the cloglog model does not converge within those groups, owing to near-zero exit rates.

The purpose of this split is to verify that the aggregate negative gradient in Table 2 is not driven by pooling heterogeneous product lines. If the negative slope reappears within each subsample, the diversification effect cannot reflect sorting of more resilient enterprises into high-count categories. Figure B.2 disaggregates the predicted exit probability itself by product subsample.

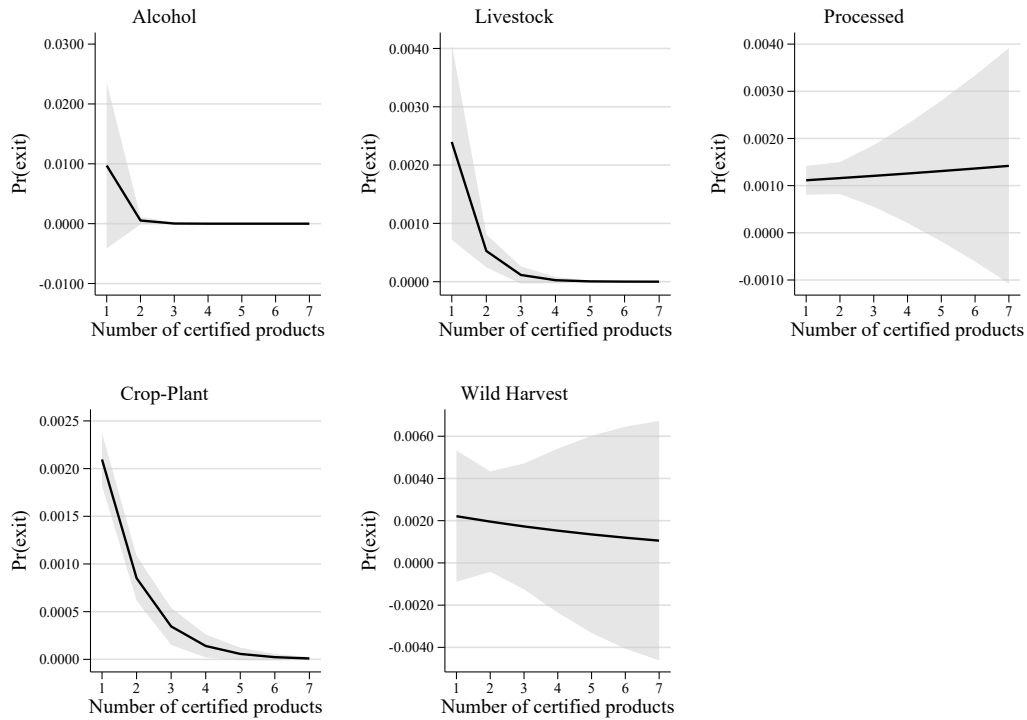
Two pieces of evidence emerge from the by-product exercise. Table B.3 reports the cloglog average marginal effect of one additional certified category, estimated separately within each subsample. The gradient is steepest for alcohol at -0.0051 ($p < 0.10$), followed by crop-plant at -0.0018 ($p < 0.01$) and livestock at -0.0015 ($p < 0.01$). Processed products ($+0.0001$) and wild harvest (-0.0006) yield flatter and imprecisely estimated effects, reflecting smaller subsample sizes and, for processed products, a high baseline level of diversification that compresses the identifying variation. In no subsample does the point estimate reverse into a meaningful positive effect.

Table B.3: Diversification effect by product subsample (cloglog average marginal effects)

	(1)	(2)	(3)	(4)	(5)
Number of certified categories	-0.0051* (0.0029)	-0.0015*** (0.0005)	0.0001 (0.0002)	-0.0018*** (0.0003)	-0.0006 (0.0023)
Observations	5,084	38,181	66,224	180,765	3,465
Product subsample	Alcohol	Livestock	Processed	Crop-Plant	Wild Harvest

Notes: This table reports the cloglog average marginal effect of the number of certified product categories, estimated separately for each product subsample. Each column restricts the sample to enterprises holding at least one certification in the named category; the subsamples therefore overlap (an enterprise certified in both crop-plant and livestock appears in both columns). The regressor is the full count of distinct certified product categories across all seven product categories, identical to the regressor in column (1) of Table 2, not a count restricted to the subsample category. Maple/forest and organic basket subsamples are omitted because the model did not converge. All specifications include region, certifier, and entry-cohort controls. Standard errors clustered at enterprise level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

**Figure B.2:** Predicted exit probability vs. number of certified product categories, by product subsample

Notes: This figure plots the predicted monthly exit probability from the cloglog model with entry-cohort fixed effects against the number of distinct certified product categories held by the enterprise, restricted to enterprises holding at least one certification in the indicated product category. Shaded bands are 95% confidence intervals, with all other covariates held at observed values. Maple/forest and organic basket subsamples are omitted because the cloglog model does not converge within those groups, consistent with the near-zero exit rates observed for those categories in Table 1.

Figure B.2 displays the same subsample estimates as non-linear predicted-probability curves rather than single marginal effects. Within the crop-plant, alcohol, and livestock subsamples the predicted monthly exit probability declines monotonically with the number of certified product categories, falling by roughly one quarter to one third as enterprises move from one to five or more activity categories. The

key implication, common to the table and the figure, is that the diversification effect survives within most product subsamples: holding the enterprise's dominant product category fixed, adding more distinct categories still lowers exit risk. This rules out a compositional interpretation under which diversification simply proxies for enterprises selecting into structurally more resilient product lines, and it reinforces the interpretation of diversification as a genuine risk-spreading mechanism against product-specific market and regulatory shocks.

B.3 Within- vs. Across-Category Diversification

Table B.4 decomposes the aggregate diversification regressor of Table 2 into two scalars that enter the hazard model jointly. The first is breadth, N_i , the number of distinct product categories in which enterprise i holds at least one certification. This is identical to the regressor in columns (2), (4), and (6) of Table 2 and, by construction, ignores within-category intensity. Let $\tilde{N}_i$ denote the total number of certified products held by enterprise i , summed across all categories (so an enterprise certified for three crop-plant products and two livestock products has $N_i = 2$ and $\tilde{N}_i = 5$). The second scalar is depth, $\tilde{N}_i - N_i$, the number of extra certified products beyond the first within each category. Depth therefore measures how many products an enterprise accumulates within the categories it has already entered, holding breadth fixed.

When constructing N_i and $\tilde{N}_i$, we count certifications only in the six protective categories (alcohol, maple/forest, livestock, processed, crop-plant, organic baskets). Wild-harvest certifications are not counted because the wild-harvest indicator carries a significantly positive exit coefficient in Table 1; counting it would mix risk-increasing variation into measures meant to capture protective diversification. The change is in how breadth and depth are defined, not in which enterprises are estimated: every enterprise in Table 2 remains in the regression, so the cloglog and logit panels still contain 423,362 rows and the Cox dataset still contains 4,100 spells. An enterprise certified only in wild harvest contributes $N_i = 0$ and $\tilde{N}_i - N_i = 0$ on these measures but is still in the sample.

Table B.4: Robustness: within- vs. across-category diversification

	Cloglog (AME) (1)	Cox (log HR) (2)	Logit (AME) (3)
<i>Duration dependence</i>			
ln(age)	0.0032*** (0.0004)		0.0032*** (0.0004)
ln(age) ²	-0.0004*** (0.0001)		-0.0004*** (0.0001)
<i>Diversification margin</i>			
Breadth: number of certified product categories	-0.0006*** (0.0002)	-0.2616** (0.1071)	-0.0006*** (0.0002)
Depth: extra items within categories	-0.0001 (0.0001)	0.0554 (0.0512)	-0.0001 (0.0001)
Observations	423,362	4,100	423,362
Region FE	Yes	Yes	Yes
Certifier FE	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Calendar-year FE	No	No	No

Notes: This table reports joint estimation of breadth and depth as defined in Section B.3. Columns (1) and (3) report cloglog and logit average marginal effects; column (2) reports Cox log hazard ratios. All specifications include region, certifier, and entry-cohort fixed effects, plus log-polynomial duration controls (cloglog and logit). Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Entering breadth and depth together lets the data say which of the two margins carries the protective diversification effect documented in the main text. The breadth coefficient matches the column-(2) estimate of Table 2 to three decimal places: the cloglog and logit average marginal effects are both -0.0006 with $p < 0.001$, and the Cox log hazard ratio is -0.262 with $p = 0.015$. An additional distinct category therefore lowers the monthly exit probability by about 0.06 percentage points on top of the baseline hazard of roughly 0.5 percent per month, or about one-eighth of the baseline.

The depth coefficient, by contrast, is small and statistically indistinguishable from zero in all three estimators: cloglog and logit average marginal effects of -0.0001 with $p = 0.44$, and a Cox log hazard ratio of $+0.055$ with $p = 0.28$. The Cox coefficient even flips from negative to a non-significant positive, ruling out a near-miss negative effect. The depth null is not a power issue. Depth has a standard deviation of roughly 1.0 items in both the enterprise-month and enterprise-spell samples, which is about twice the standard deviation of breadth (0.42–0.49), so the identifying variation in depth is ample. Nor is the null driven by wild-harvest contamination, since the wild-harvest category has already been netted out of the scalar.

On average, then, protection operates on the breadth margin: broadening the set of certified activities hedges exposure to independent, category-specific shocks, whereas deepening within an already-certified activity leaves the enterprise on the same risk profile. This finding reconciles naturally with the category-level evidence in Table B.7 below. There, the “2+” coefficient is more strongly negative than the “1” coefficient for maple/forest, livestock, and processed products, indicating that depth does provide additional protection within those categories; but it is close to zero or positive for crop-plant and wild harvest. Averaging across categories yields no detectable aggregate depth effect, which is exactly the pattern that a single scalar depth regressor is bound to deliver when the category-specific depth returns

are heterogeneous in this way. The two tables are therefore consistent rather than contradictory, and together they support a within-enterprise insurance mechanism in which the protective margin comes from broadening across activities rather than from deepening within any one of them.

B.4 Wild-Harvest Control in the Restricted Diversification Count

Columns (2), (4), and (6) of Table 2 restrict the category count to the six protective categories, excluding wild harvest from the diversification measure. Because that specification does not separately control for wild-harvest status, the estimated diversification coefficient could partly capture the direct association between wild-harvest certification and exit documented in Table 1, if wild-harvest certification is correlated with the restricted category count. Table B.5 therefore re-estimates the restricted-count specification with a wild-harvest indicator included as an additional control.

The diversification coefficient is essentially unchanged. The cloglog and logit average marginal effects are identical to four decimal places (-0.0006 , $p < 0.01$), while the Cox log hazard ratio changes only from -0.225 to -0.207 , with both estimates significant at the 5% level. Wild-harvest status itself enters positively and significantly ($+0.0008$, $p < 0.05$, in the cloglog and logit specifications; $+0.704$, $p < 0.01$, in the Cox specification), consistent in sign and significance with its coefficient in Table 1. The restricted-count diversification effect is therefore robust to controlling explicitly for wild-harvest status.

Table B.5: Robustness: restricted diversification count with explicit wild-harvest control

	Cloglog (AME)	Cox (log HR)	Logit (AME)
Breadth: number of certified product categories	-0.0006*** (0.0001)	-0.2067** (0.0949)	-0.0006*** (0.0001)
Wild harvest	0.0008** (0.0004)	0.7044*** (0.2528)	0.0008** (0.0004)
Observations	423,362	4,100	423,362
Region FE	Yes	Yes	Yes
Certifier FE	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Calendar-year FE	No	No	No

Notes: This table re-estimates columns (2), (4), and (6) of Table 2 (breadth, count of the six protective categories excluding wild harvest) with a wild-harvest indicator entered as an explicit additional regressor, so wild-harvest status is controlled for directly rather than only removed from the count. Columns report cloglog and logit average marginal effects and Cox log hazard ratios. All specifications include region, certifier, and entry-cohort fixed effects, plus log-polynomial duration controls (cloglog and logit). Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.5 Raw Category Counts

Table B.6 replaces the seven binary product indicators of Table 1 with their raw counts, so each covariate is the number of certified products an enterprise holds in that category rather than an indicator for having any. The specification is otherwise identical to columns (1), (3), and (5) of Table 1: estimation is by cloglog (column 1), Cox partial likelihood (column 2), and logit (column 3), all with region, certifier, and entry-cohort fixed effects and enterprise-clustered standard errors.

This check has two purposes. First, it verifies that the main-text product-type effects are not driven by treating one certified product the same as five within a given category. Second, it provides the

category-by-category analogue of the aggregate within-enterprise depth margin examined jointly with breadth in Table B.4: here each category-specific coefficient measures the marginal risk of adding one more product within its own category, holding the other categories fixed.

The pattern of signs in Table B.6 mirrors the binary-indicator results of Table 1. Every protective category retains a negative and statistically significant coefficient once the indicator is replaced by a raw count: maple/forest products at -0.0014 ($p < 0.01$), livestock at -0.0010 ($p < 0.01$), processed products at -0.0005 ($p < 0.01$), organic baskets at -0.0021 ($p < 0.10$), alcoholic beverages at -0.0009 ($p < 0.10$), and crop/plant products at -0.0001 ($p < 0.05$). Wild harvest remains the single risk-increasing category, with a coefficient of $+0.0006$ ($p < 0.05$). The economic magnitudes also scale as expected: an extra certified product of maple/forest, livestock, or organic baskets carries a larger marginal protection than an extra crop/plant or processed product, consistent with the binary-indicator magnitudes in Table 1.

Table B.6: Robustness: raw category counts

	Cloglog (AME) (1)	Cox (log HR) (2)	Logit (AME) (3)
<i>Duration dependence</i>			
$\ln(\text{age})$	0.0032*** (0.0004)		0.0032*** (0.0004)
$\ln(\text{age})^2$	-0.0004*** (0.0001)		-0.0004*** (0.0001)
<i>Activity counts (certified units per type)</i>			
Alcoholic beverages	-0.0009* (0.0005)	-0.5781* (0.3241)	-0.0009* (0.0005)
Maple syrup / forest products	-0.0014*** (0.0002)	-0.8900*** (0.1300)	-0.0014*** (0.0002)
Livestock	-0.0010*** (0.0002)	-0.3161** (0.1360)	-0.0010*** (0.0002)
Processed products	-0.0005*** (0.0001)	-0.1209* (0.0722)	-0.0005*** (0.0001)
Crop / plant products	-0.0001** (0.0001)	-0.0755 (0.0479)	-0.0001** (0.0001)
Organic baskets	-0.0021* (0.0012)	-1.2716* (0.7007)	-0.0021* (0.0012)
Wild harvest	0.0006** (0.0002)	0.4348* (0.2231)	0.0006** (0.0002)
Observations	423,362	4,100	423,362
Region FE	Yes	Yes	Yes
Certifier FE	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Calendar-year FE	No	No	No

Notes: This table shows average marginal effects in the cloglog and logit columns and log hazard ratios in the Cox columns. Activity counts are continuous (number of certified products per category). All specifications include region, certifier, and entry-cohort fixed effects. Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.6 Grouped Category Counts (0, 1, 2+)

Table B.7 relaxes the linearity assumption of Table B.6 by grouping the raw count within each category into three bins: zero certified products (the omitted base), exactly one product, or two or more products. Each category therefore contributes up to two coefficients.

The economic content of the table is concentrated in the “2+” coefficient, which isolates the marginal effect of deepening within an already-certified category: moving from one product to two or more products within the same category. This is the category-level counterpart of the aggregate depth scalar in Table B.4.

The category-by-category comparison tells a clean story. In the three categories with the strongest “1” protective effect (maple/forest, livestock, and processed products), the “2+” coefficient is at least as large in magnitude, indicating that holding a second certified product within the category provides additional protection over and above the first: maple/forest moves from -0.0010 at one product to -0.0014 at two or more, livestock from -0.0004 to -0.0014 , and processed products from -0.0006 to -0.0010 , with all six coefficients significant at the 1 or 5 percent level. In crop/plant, the “1” coefficient is actually positive ($+0.0005$, $p < 0.10$) while the “2+” coefficient is negative and indistinguishable from zero, so depth within crop/plant does not add protection. Wild harvest shows a similar pattern in reverse: “1” is strongly positive ($+0.0014$, $p < 0.05$), capturing the risk-increasing role of the category documented in Table 1, while “2+” is essentially zero.

Viewed together with Table B.4, this reconciles the aggregate depth null with the category-level heterogeneity. Depth pays off within the three protective categories where enterprises are naturally resilient, contributes little in crop/plant, and is essentially irrelevant once wild-harvest enterprises have passed the single-product threshold. A single scalar depth regressor that averages across all seven categories is bound to net these opposing contributions to zero, which is precisely what Table B.4 shows. The two tables are therefore consistent, not contradictory, and they jointly support a within-enterprise insurance mechanism in which the protective value of an extra product depends on the category to which it is added.

Table B.7: Robustness: grouped category counts (0, 1, 2+)

	Cloglog (AME) (1)	Cox (log HR) (2)	Logit (AME) (3)
<i>Duration dependence</i>			
ln(age)	0.0032*** (0.0004)		0.0032*** (0.0004)
ln(age) ²	-0.0004*** (0.0001)		-0.0004*** (0.0001)
<i>Alcoholic beverages (ref: 0)</i>			
1	-0.0008*** (0.0003)	-0.6318** (0.3223)	-0.0008*** (0.0003)
<i>Maple syrup / forest (ref: 0)</i>			
1	-0.0010*** (0.0002)	-0.8384*** (0.1582)	-0.0010*** (0.0002)
2+	-0.0014** (0.0006)	-1.0390 (1.0389)	-0.0014** (0.0006)
<i>Livestock (ref: 0)</i>			
1	-0.0004* (0.0002)	-0.0908 (0.2005)	-0.0004* (0.0002)
2+	-0.0014*** (0.0001)	-1.1667** (0.4881)	-0.0014*** (0.0001)
<i>Processed products (ref: 0)</i>			
1	-0.0006*** (0.0002)	-0.2221 (0.1516)	-0.0006*** (0.0002)
2+	-0.0010*** (0.0002)	-0.3363 (0.2171)	-0.0010*** (0.0002)
<i>Crop / plant (ref: 0)</i>			
1	0.0005* (0.0003)	0.1262 (0.1457)	0.0005* (0.0003)
2+	-0.0003 (0.0002)	-0.2099 (0.1446)	-0.0003 (0.0002)
<i>Organic baskets (ref: 0)</i>			
1	-0.0011*** (0.0003)	-1.2007* (0.6850)	-0.0011*** (0.0003)
<i>Wild harvest (ref: 0)</i>			
1	0.0014** (0.0007)	0.7860*** (0.2634)	0.0014** (0.0007)
2+	0.0001 (0.0014)	-0.3321 (1.1049)	0.0001 (0.0014)
Observations	423,362	4,100	423,362
Region FE	Yes	Yes	Yes
Certifier FE	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Calendar-year FE	No	No	No

Notes: This table shows average marginal effects in the cloglog and logit columns and log hazard ratios in the Cox columns. Activity counts are grouped as 0, 1, or 2+ certified products per category (base: 0). All specifications include region, certifier, and entry-cohort fixed effects. Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.7 New Entrants Only

Table B.8 re-estimates the main specification on the subsample of enterprises whose first organic certification falls within the 2023–2025 observation window. This restriction eliminates left-truncation bias: enterprises that entered before 2023 contribute no information about their early-age hazard to the baseline estimator, and the log-polynomial duration term is identified only from within-sample certification spells.

Table B.8: Robustness: new entrants only

	Cloglog (AME) (1)	Cox (log HR) (2)	Logit (AME) (3)
<i>Duration dependence</i>			
ln(age)	0.0680*** (0.0152)		0.0678*** (0.0150)
ln(age) ²	-0.0137*** (0.0031)		-0.0137*** (0.0031)
<i>Certified product type</i>			
Alcoholic beverages	-0.0013 (0.0034)	-0.3498 (0.6552)	-0.0013 (0.0034)
Maple syrup / forest products	-0.0053** (0.0025)	-1.0551** (0.4741)	-0.0053** (0.0025)
Livestock	0.0006 (0.0048)	0.0877 (0.9755)	0.0006 (0.0048)
Processed products	0.0008 (0.0027)	0.1398 (0.5323)	0.0008 (0.0027)
Crop / plant products	-0.0019 (0.0022)	-0.3717 (0.4325)	-0.0019 (0.0022)
Observations	15,154	662	15,154
Region FE	Yes	Yes	Yes
Certifier FE	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Calendar-year FE	No	No	No

Notes: This table shows estimates from the sample restricted to enterprises entering certification within the observation window (no left-truncation), with average marginal effects in the cloglog and logit columns and log hazard ratios in the Cox columns. The organic-baskets and wild-harvest indicators remain in the regression specification but are omitted from the displayed output because they are unidentified in this subsample (near-zero exit cells yield zero-valued average marginal effects and undefined standard errors). All specifications include region, certifier, and entry-cohort fixed effects. Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The inverted-U duration profile is preserved under the left-truncation restriction: the ln(age) and ln(age)² average marginal effects reported in Table B.8 are +0.068 and −0.014, both significant at the 1 percent level, matching the sign pattern of the main specification. The trade-off of the restriction is a much smaller sample: 662 enterprises, 15,556 enterprise-months, and 76 observed exits, roughly one-sixth of the main panel by enterprise count and about one-ninth of the 668 exits used in the main estimation. The maple/forest protective effect, which is identified by genuine within-sample exits among 2023–2025 entrants, persists at the 5 percent level in both the cloglog and Cox specifications. The remaining product-type coefficients lose statistical significance but retain the same sign as in Table 1 (negative for alcohol and crop-plant, essentially zero for livestock and processed, whose sign flips are

economically indistinguishable from zero). The loss of precision is expected: the main-text estimates are identified largely by exits at ages of 30 to 60 months, which cannot occur in a subsample of enterprises first certified after 2023. The exercise therefore verifies that the main-text findings are not driven by left-truncation, not that every product-type effect remains precisely estimated on one sixth of the data. The coefficients for organic baskets and wild harvest are not reported in the table because near-zero exit cells in this subsample leave them unidentified; the indicators remain in the regression specification so that the other coefficients retain their conditional interpretation.

B.8 Age-Band Duration Controls vs. Log-Polynomial

The main-text specification captures duration dependence with a log-polynomial $\alpha(t) = \beta_1 \ln(\text{age}) + \beta_2 \ln^2(\text{age})$. The estimated coefficients, $\beta_1 \approx 2.04$ and $\beta_2 \approx -0.27$, generate an inverted-U hazard profile that peaks near 43 months of certification age. The polynomial has a built-in single-peaked shape, raising the question whether the inverted-U is a genuine feature of the data or a pattern mechanically imposed by the functional form. A related concern is that a mis-specified duration term can contaminate the other coefficients of interest, since the product-type and diversification effects are identified net of the duration trend.

Table B.9 settles both concerns with a falsification exercise. Columns (1)–(2) replace the polynomial with non-parametric step-function age-band indicators (1–12 months is the reference, followed by 13–24, 25–36, 37–48, 49–60, and 61+ months). Step functions impose no shape on the hazard: whatever profile emerges is whatever the data support. Columns (3)–(4) re-estimate the baseline log-polynomial specification on the same sample. If the polynomial were mechanically driving either the inverted-U or the downstream covariate estimates, the age-band and log-polynomial specifications would give different answers for those covariates. They do not.

Two features of the table are worth emphasising. First, the age-band coefficients themselves trace the inverted-U non-parametrically. The average marginal effect rises monotonically from +0.0007 at 13–24 months to a plateau of +0.0012 over 37–48 and 49–60 months, then drops back to +0.0005 at 61 months and beyond, all significant at the 1 percent level. The step-function peak at roughly three to five years of certification age is consistent with the 43-month turning point implied by the log-polynomial, and it rules out the interpretation that the inverted-U is a by-product of smoothing. Second, the product-type and diversification coefficients in columns (1)–(2) match those in columns (3)–(4) essentially to the fourth decimal place: the maple/forest AME is -0.0012 under either specification, livestock is -0.0011 , processed products is -0.0010 versus -0.0009 , wild harvest is $+0.0010$ versus $+0.0009$, and the alcohol, crop/plant, and organic-basket coefficients are numerically identical across the two duration specifications. The product-type effects reported in the main text are therefore not driven by the duration functional form, and the non-parametric step functions corroborate both the inverted-U hazard profile and the product-type findings.

Table B.9: Robustness: age-band duration controls vs. log-polynomial

	Age-band duration		ln-polynomial duration	
	(1)	(2)	(3)	(4)
<i>Duration dependence (ref: 1–12 months)</i>				
13–24 months	0.0007*** (0.0002)	0.0007*** (0.0002)		
25–36 months	0.0010*** (0.0003)	0.0010*** (0.0003)		
37–48 months	0.0012*** (0.0003)	0.0012*** (0.0003)		
49–60 months	0.0012*** (0.0003)	0.0012*** (0.0003)		
61+ months	0.0005*** (0.0002)	0.0005*** (0.0002)		
ln(age)			0.0032*** (0.0004)	0.0032*** (0.0004)
ln(age) ²			-0.0004*** (0.0001)	-0.0004*** (0.0001)
<i>Certified product type</i>				
Alcoholic beverages	-0.0009* (0.0005)	-0.0009* (0.0005)	-0.0009* (0.0005)	-0.0009* (0.0005)
Maple syrup / forest products	-0.0012*** (0.0003)	-0.0012*** (0.0003)	-0.0012*** (0.0003)	-0.0012*** (0.0003)
Livestock	-0.0011*** (0.0003)	-0.0011*** (0.0003)	-0.0011*** (0.0003)	-0.0011*** (0.0003)
Processed products	-0.0010*** (0.0002)	-0.0010*** (0.0002)	-0.0009*** (0.0002)	-0.0009*** (0.0002)
Crop / plant products	-0.0001 (0.0002)	-0.0001 (0.0002)	-0.0000 (0.0002)	-0.0000 (0.0002)
Organic baskets	-0.0022* (0.0012)	-0.0022* (0.0012)	-0.0022* (0.0012)	-0.0022* (0.0012)
Wild harvest	0.0010*** (0.0004)	0.0010*** (0.0004)	0.0009** (0.0004)	0.0009** (0.0004)
Observations	423,362	423,362	423,362	423,362
Region FE	Yes	Yes	Yes	Yes
Certifier FE	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes
Calendar-year FE	No	No	No	No

Notes: This table shows average marginal effects in all columns. Columns (1)–(2) model duration dependence via age-band dummies (base: 1–12 months); columns (3)–(4) use the $\ln(\text{age}) + \ln^2(\text{age})$ polynomial (baseline specification). All specifications include region, certifier, and entry-cohort fixed effects. Standard errors clustered at enterprise level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Proof of Proposition 1

We prove Proposition 1: under conditions (i) $\pi_o^*(\sigma_N) < \pi_c^*$ and (ii) $\omega^\top \pi_o^* > \pi_c^*$, and at the calibrated $(\hat{\Omega}, \hat{\pi}_o^*, \hat{\pi}_c^*)$ of Table 3, the aggregate exit probability P_t^θ is strictly decreasing in β_θ . The argument proceeds in two steps: the state-conditional derivative is signed via Neumann expansion, then the

aggregate inherits the sign from a β_θ -invariant weighted sum. The three building blocks are:

$$V_C^\theta = \frac{1}{1 - \beta_\theta} \pi_c^*, \quad (\text{C.1})$$

$$\mathbf{V}_O^\theta = (\mathbb{I}_N - \beta_\theta \Omega)^{-1} \boldsymbol{\pi}_o^*, \quad (\text{C.2})$$

$$P_{t,\sigma_j}^\theta = \frac{1}{1 + \exp[-\gamma(V_C^\theta - V_O^\theta(\sigma_j))]}, \quad (\text{C.3})$$

where $\boldsymbol{\pi}_o^* = [\pi_o^*(\sigma_1), \dots, \pi_o^*(\sigma_N)]^\top$ collects maximized per-period organic profits across pest states, π_c^* is the maximized conventional profit, and $\Omega \in \mathbb{R}^{N \times N}$ is row-stochastic and primitive.

Differentiation of the logit. Define the value gap $\Delta_j \equiv V_C^\theta - V_O^\theta(\sigma_j)$, so that $P_{t,\sigma_j}^\theta = \Lambda(\gamma \Delta_j)$ where $\Lambda(x) \equiv 1/(1 + e^{-x})$ denotes the logistic function. Since $\Lambda'(x) = \Lambda(x)(1 - \Lambda(x))$, the chain rule gives

$$\frac{\partial P_{t,\sigma_j}^\theta}{\partial \beta_\theta} = \gamma P_{t,\sigma_j}^\theta (1 - P_{t,\sigma_j}^\theta) \left(\frac{\partial V_C^\theta}{\partial \beta_\theta} - \frac{\partial V_O^\theta(\sigma_j)}{\partial \beta_\theta} \right).$$

Because $\gamma > 0$ and $P_{t,\sigma_j}^\theta \in (0, 1)$, the prefactor $\gamma P_{t,\sigma_j}^\theta (1 - P_{t,\sigma_j}^\theta)$ is strictly positive, so

$$\text{sgn} \left(\frac{\partial P_{t,\sigma_j}^\theta}{\partial \beta_\theta} \right) = \text{sgn} \left(\frac{\partial V_C^\theta}{\partial \beta_\theta} - \frac{\partial V_O^\theta(\sigma_j)}{\partial \beta_\theta} \right).$$

It remains to compute each derivative and compare them term by term.

Derivative of the conventional value. From Equation (C.1), direct differentiation gives

$$\frac{\partial V_C^\theta}{\partial \beta_\theta} = \frac{\pi_c^*}{(1 - \beta_\theta)^2}.$$

Applying the power-series identity $\sum_{t=1}^{\infty} t x^{t-1} = 1/(1 - x)^2$ with $x = \beta_\theta$:

$$\frac{\partial V_C^\theta}{\partial \beta_\theta} = \sum_{t=1}^{\infty} t \beta_\theta^{t-1} \pi_c^*. \quad (\text{C.4})$$

Each term $t \beta_\theta^{t-1} \pi_c^*$ is the marginal weight placed on the conventional profit flow at horizon t when patience rises by one unit.

Derivative of the organic value by implicit differentiation. The vector Bellman system reads $(\mathbb{I}_N - \beta_\theta \Omega) \mathbf{V}_O^\theta = \boldsymbol{\pi}_o^*$. Differentiating both sides with respect to β_θ :

$$\frac{d}{d\beta_\theta} (\mathbb{I}_N - \beta_\theta \Omega) \mathbf{V}_O^\theta + (\mathbb{I}_N - \beta_\theta \Omega) \frac{d\mathbf{V}_O^\theta}{d\beta_\theta} = \mathbf{0}.$$

Since $\frac{d}{d\beta_\theta} (\mathbb{I}_N - \beta_\theta \Omega) = -\Omega$, this becomes

$$-\Omega \mathbf{V}_O^\theta + (\mathbb{I}_N - \beta_\theta \Omega) \frac{d\mathbf{V}_O^\theta}{d\beta_\theta} = \mathbf{0}.$$

Premultiplying by $(\mathbb{I}_N - \beta_\theta \Omega)^{-1}$, which exists because Ω is row-stochastic so $\rho(\beta_\theta \Omega) = \beta_\theta \rho(\Omega) = \beta_\theta < 1$, where $\rho(\cdot)$ denotes the spectral radius:

$$\frac{d\mathbf{V}_O^\theta}{d\beta_\theta} = (\mathbb{I}_N - \beta_\theta \Omega)^{-1} \Omega \mathbf{V}_O^\theta. \quad (\text{C.5})$$

The j -th scalar component is $\frac{\partial V_O^\theta(\sigma_j)}{\partial \beta_\theta} = \mathbf{e}_j^\top (\mathbb{I}_N - \beta_\theta \Omega)^{-1} \Omega \mathbf{V}_O^\theta$, where $\mathbf{e}_j$ is the j -th standard unit vector.

Neumann-series expansion. Equation (C.5) expresses $\partial \mathbf{V}_O^\theta / \partial \beta_\theta$ as a matrix-vector product, which is not directly comparable to the scalar expression for $\partial V_C^\theta / \partial \beta_\theta$ in Equation (C.4). We convert it to scalar form by expanding the matrix inverse as a Neumann series and reading off the j -th component. Because $\rho(\beta_\theta \Omega) < 1$, the matrix inverse admits the convergent power-series expansion

$$(\mathbb{I}_N - \beta_\theta \Omega)^{-1} = \sum_{t=0}^{\infty} \beta_\theta^t \Omega^t.$$

Substituting into the closed-form $\mathbf{V}_O^\theta = (\mathbb{I}_N - \beta_\theta \Omega)^{-1} \pi_o^*$ and taking the j -th component yields

$$V_O^\theta(\sigma_j) = \sum_{t=0}^{\infty} \beta_\theta^t [\Omega^t \pi_o^*]_j = \sum_{t=0}^{\infty} \beta_\theta^t \mathbb{E}_{\sigma_j} [\pi_o^*(\sigma^t)],$$

where the second equality uses the fact that the j -th row of Ω^t is the t -step transition distribution from σ_j , so its inner product with π_o^* is the conditional expectation of $\pi_o^*(\sigma^t)$ given the starting state $\sigma^0 = \sigma_j$. The $t = 0$ term is the deterministic current-period flow $\pi_o^*(\sigma_j)$; subsequent terms are expected discounted future flows under the Markov dynamics.

Differentiating term by term with respect to β_θ (the conditional expectation depends on Ω but not on β_θ) gives

$$\frac{\partial V_O^\theta(\sigma_j)}{\partial \beta_\theta} = \sum_{t=1}^{\infty} t \beta_\theta^{t-1} \mathbb{E}_{\sigma_j} [\pi_o^*(\sigma^t)], \quad (\text{C.6})$$

where the $t = 0$ term vanishes because $\partial \beta_\theta^0 / \partial \beta_\theta = 0$. The right-hand side is now directly comparable to Equation (C.4).

Term-by-term comparison. Subtracting (C.4) from (C.6) defines the function

$$G(\sigma_j, \beta_\theta) \equiv \frac{\partial V_O^\theta(\sigma_j)}{\partial \beta_\theta} - \frac{\partial V_C^\theta}{\partial \beta_\theta} = \sum_{t=1}^{\infty} t \beta_\theta^{t-1} \delta_t(\sigma_j), \quad \delta_t(\sigma_j) \equiv \mathbb{E}_{\sigma_j} [\pi_o^*(\sigma^t)] - \pi_c^*. \quad (\text{C.7})$$

The weights $t \beta_\theta^{t-1}$ are strictly positive for all $t \geq 1$, so $G(\sigma_j, \beta_\theta) > 0$, and hence $\partial P_{t, \sigma_j}^\theta / \partial \beta_\theta < 0$, whenever $\delta_t(\sigma_j) \geq 0$ for all t , with strict inequality for at least one term.

Sign of G at the calibrated parameters. $G(\sigma_j, \beta_\theta)$ does not reduce to a term-by-term sign condition under the economic assumptions of Proposition 1. Under (i), $\pi_o^*(\sigma_N) < \pi_c^*$ implies early-horizon $\delta_t(\sigma_j)$ can be negative when the chain is likely to visit σ_N . Under (ii), ergodicity of Ω gives $[\Omega^t]_{j, \cdot} \rightarrow \omega^\top$ geometrically, so $\delta_t(\sigma_j) \rightarrow \omega^\top \pi_o^* - \pi_c^* > 0$. Whether the late-horizon positive tail dominates the early-horizon transient shortfalls is a quantitative question, settled by direct evaluation of G .

We verify the sign directly at the calibrated primitives. With $\hat{\Omega}$ from Table 3, the stationary distribution is $\hat{\omega} = (0.046, 0.196, 0.758)^\top$. Averaging the three crops of the calibration sample (CRAAQ corn, soybean, barley) using the same pooled prices, yields, and costs that enter the calibration, and netting the \$20/ha certification fee, gives $\hat{\pi}_o^* = (\$2,187, \$1,844, \$645)$ per hectare and $\hat{\pi}_c^* = \$824$. Conditions (i) and (ii) both hold: $\hat{\pi}_o^*(\sigma_3) = 645 < 824 = \hat{\pi}_c^*$, and $\hat{\omega}^\top \hat{\pi}_o^* = 951 > 824$. Truncating the infinite sum at $t = 1,000$ (tail bounded by $t\hat{\beta}_\theta^{t-1} \max_j |\delta_t(\sigma_j)| < 10^{-10}$ for both $\hat{\beta}_\theta$ values) and cross-checked against the closed form $(\mathbb{I}_N - \beta_\theta \hat{\Omega})^{-1} \hat{\Omega} \mathbf{V}_O^\theta - \hat{\pi}_c^*/(1 - \beta_\theta)^2$, the values are reported in Table C.1.

Table C.1: $G(\sigma_j, \beta_\theta)$ at the calibrated primitives (dollars per hectare).

	$\hat{\beta}_\theta = 0.90$	$\hat{\beta}_{\bar{\theta}} = 0.97$
σ_1 (good)	+15,452	+144,927
σ_2 (moderate)	+13,762	+143,031
σ_3 (severe)	+12,352	+141,426

Notes: This table presents numerical values of $G(\sigma_j, \beta_\theta) = \partial V_O^\theta(\sigma_j)/\partial \beta_\theta - \partial V_C^\theta/\partial \beta_\theta$, evaluated at the calibrated primitives $(\hat{\Omega}, \hat{\pi}_o^*, \hat{\pi}_c^*)$ from Table 3. Each row corresponds to a starting pest state σ_j ($j = 1, 2, 3$); the two columns report values at the calibrated impatient and patient discount factors. All six entries are strictly positive (in dollars per hectare), confirming $\partial P_{t,\sigma_j}^\theta/\partial \beta_\theta < 0$ in every state and for both producer types. Values obtained by truncating the infinite sum at $t = 1,000$ and cross-checked against the closed-form $(\mathbb{I}_N - \beta_\theta \hat{\Omega})^{-1} \hat{\Omega} \mathbf{V}_O^\theta - \hat{\pi}_c^*/(1 - \beta_\theta)^2$.

All six values are strictly positive, so $\partial V_O^\theta(\sigma_j)/\partial \beta_\theta > \partial V_C^\theta/\partial \beta_\theta$ at both calibrated β values and for every starting state σ_j . Collecting the above:

$$\frac{\partial P_{t,\sigma_j}^\theta}{\partial \beta_\theta} = \underbrace{-\gamma P_{t,\sigma_j}^\theta (1 - P_{t,\sigma_j}^\theta)}_{>0} \cdot \underbrace{\sum_{t=1}^{\infty} t \beta_\theta^{t-1} \delta_t(\sigma_j)}_{>0} < 0.$$

Patient producers (larger β_θ) therefore have lower P_{t,σ_j}^θ than impatient producers in every pest state σ_j . The pest-state distribution $\Phi_{j,t}$ evolves under Ω alone and is therefore independent of β_θ . Since $P_t^\theta = \sum_j \Phi_{j,t} P_{t,\sigma_j}^\theta$ is a non-negative weighted sum with β_θ -invariant weights satisfying $\sum_j \Phi_{j,t} = 1$, the aggregate exit probability P_t^θ is also strictly decreasing in β_θ at the calibrated primitives, so $P_t^\theta > P_t^{\bar{\theta}}$. This completes the proof of Proposition 1. ■

D Calibration Inputs and Parameter Bounds

This appendix reports the production-parameter data from CRAAQ and the economically admissible parameter bounds used in the calibration of Section 4. The revenue and expense table pins down the flow profit components in Equations (3.1)–(3.2); the bounds table documents the literature anchors underlying each parameter range and is referenced in the calibration description.

Table D.1: Revenue and Expenses by Crop Management in Quebec (2019–2024)

	Corn		Soybean		Barley	
	Org.	Conv.	Org.	Conv.	Org.	Conv.
Prices (\$/t)	490	277	1151	564	458	328
Yields (t/ha)	9.1	10	2.9	3.09	2.69	4
Total costs excl. pesticides (\$/ha)	1848	1857	812	869	997	1219
Pesticide expenditures (\$/ha)	–	90	–	95	–	57

Notes: This table shows prices, yields, and production costs for Quebec field crops between 2019 and 2024, by crop and management regime, as reported in the CRAAQ *Références économiques*. These are prospective enterprise budgets constructed from farm accounting data collected by the *Groupes conseils agricoles* and validated through technical panels of agronomists and experienced producers; they describe the well-managed representative farm under normal conditions and exclude realised climate- and pest-risk losses. Total costs exclude pesticide expenditures, which are shown separately. Pesticide expenditures refer to synthetic products, absent in the organic regime. *Source:* CRAAQ.

Table D.2: Parameter Bounds Used in the Calibration

Parameter	Description	Lower	Upper	Source
β_θ	Impatient discount factor	0.70	0.90	Duquette et al. (2012)
$\beta_{\bar{\theta}}$	Patient discount factor	0.90	0.99	Grijalva et al. (2018)
γ	Logit sensitivity (\$/ha) ⁻¹	10 ⁻⁶	0.01	See notes
S_0^θ	Initial share of impatient enterprises	0	1	Free
σ_2	Moderate shock (retained yield)	0.60	0.90	Seufert et al. (2012) : moderate loss 10–40%
σ_3	Severe shock (retained yield)	0.35	0.55	Seufert et al. (2012)
Ω_{ij}	Transition probabilities	0	1	Row-sum constraint

Note: Discount factors map to annual rates via $r = 1/\beta - 1$: the impatient interval [0.70, 0.90] corresponds to $r \in [11\%, 43\%]$, bracketing the experimental rates of 28–43% reported for U.S. farmers by [Duquette et al. \(2012\)](#); the patient interval [0.90, 0.99] corresponds to $r \in [1\%, 11\%]$, containing the long-horizon range of 2–6% estimated by [Grijalva et al. \(2018\)](#) on a 20-year elicitation. $\sigma_1 = 1$ is normalised. The logit sensitivity γ is bounded in $[10^{-6}, 10^{-2}]$ (\$/ha)⁻¹ to span the non-degenerate range between the limiting cases discussed in Section 3: at $\gamma = 10^{-6}$ the logit is essentially flat against typical \$100–\$1,000/ha value gaps, so exit is driven almost entirely by idiosyncratic noise; at $\gamma = 10^{-2}$ the logit argument reaches 1–10, a responsive but not deterministic choice rule. All Ω_{ij} are bounded in $[0, 1]$ subject to row-sum constraints. We impose $\Omega_{13} = \Omega_{31} = 0$: pest pressure evolves gradually, ruling out direct transitions between the most favourable and most adverse states in a single period. This tridiagonal (banded) structure reflects the biology governing pest and disease abundance in field-crop systems. Weed seedbanks persist for multiple seasons and deplete only through several years of cumulative rotation and cultural practices ([Borgy et al. 2015](#)), so a pest-free field cannot become severely infested within a single growing season absent a catastrophic event. Soil-borne pathogen inoculum and beneficial microbial communities similarly shift over multi-year horizons rather than discretely ([Larkin 2015](#)). Encoding this gradualism as a banded Markov chain reduces the number of free entries in Ω from six to four and tightens identification.

Figure D.1 reports the raw observed exit hazard $\hat{P}_t = d_t/n_t$ on the crop-plant subsample used as the calibration target in Section 4.1. Within the calibration window ($t \leq 20$), $\hat{P}_t$ traces a clean inverted-U: the hazard rises from roughly 0.020 at year 1, peaks within the early-tenure region consistent with the cloglog log-polynomial estimate, then declines smoothly toward about 0.005 by year 19, and 94%

of observed exits are realised by year 20. Beyond the cutoff, the discrete hazard becomes erratic as the risk set thins through right-censoring, and the wide pointwise confidence intervals across the post-window points reflect this small-sample volatility. Exits at these long durations are also increasingly driven by demographic factors outside the model: enterprises observed at $t > 20$ are by construction operated by farmers at least twenty years older than at first certification, so owner retirement, death, or intergenerational transfer become first-order drivers of exit, displacing the patience-and-pest mechanism modelled in Section 3. These two considerations together motivate the calibration cutoff $T = 20$ years.

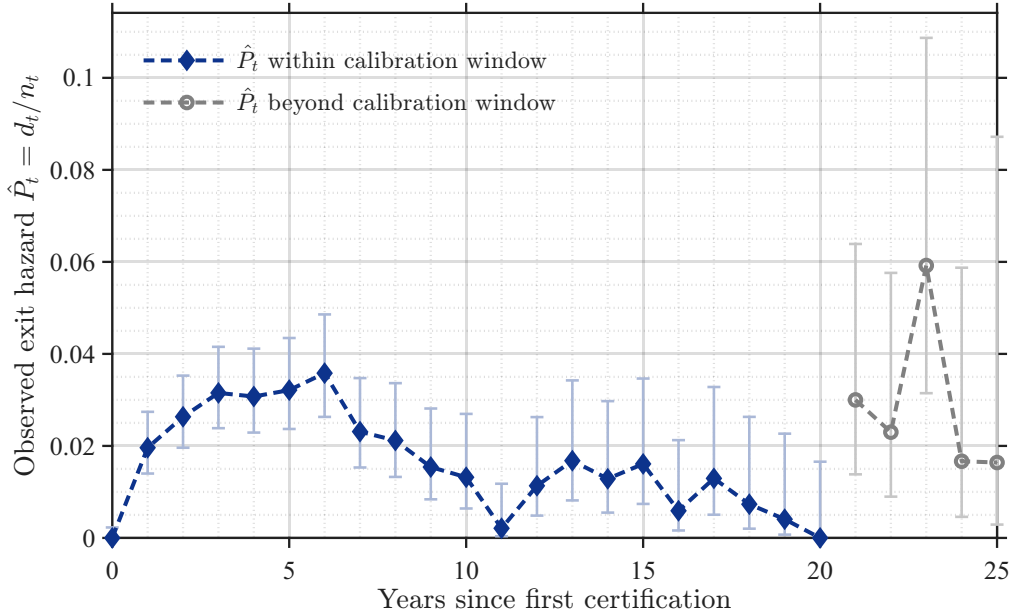


Figure D.1: Raw observed exit hazard $\hat{P}_t = d_t/n_t$ on the calibration sample

Notes: This figure shows the raw observed exit hazard $\hat{P}_t = d_t/n_t$ on the crop-plant subsample ($n_0 = 1,683$) used as the calibration target in Section 4.1, plotted against years since first certification with Wilson 95% binomial confidence bars. Dark-blue filled diamonds and dashed line correspond to durations within the calibration window ($t \leq 20$); gray hollow circles and dashed line correspond to durations beyond it ($t > 20$).

D.1 Empirical Validation of the Patience Mechanism

A central limitation of the structural model is that time preferences are not directly observable in the data: the *Portail Bio Québec* registry records certification status, product categories, and exit events, but contains no information on producers' discount factors or intertemporal attitudes. Discount factors are therefore calibrated to aggregate survival moments rather than identified from individual preference data. To provide empirical grounding for the patience mechanism despite this constraint, we exploit certification in maple syrup or livestock as a proxy for patience.²⁶ Under the model's interpretation, patient enterprises discount future organic rents less steeply and therefore delay exit; the proxy should predict exit age positively and exit hazard negatively, conditional on the controls in the main cloglog specification.

²⁶Maple orchards require years of tree development before tapping is viable; livestock operations involve breeding cycles that only pay off over multi-year horizons. Both activities therefore require a willingness to delay financial returns that distinguishes forward-looking producers from those with short planning horizons.

Table D.3 reports cloglog estimates for the baseline enterprise-month sample. Column (1) includes the patience proxy alongside the log-polynomial duration controls; columns (2)–(4) add product-category dummies, certifier and region fixed effects, and entry-cohort fixed effects in sequence. The coefficient on the long-horizon proxy is negative and statistically significant across all four specifications: enterprises with maple or livestock certification face meaningfully lower conditional exit hazard than comparable enterprises holding only short-horizon product categories.

Table D.3: Cloglog estimates: long-horizon proxy and exit hazard

	(1)	(2)	(3)	(4)
Long-horizon proxy	-0.629*** (0.083)	-0.692*** (0.102)	-0.692*** (0.116)	-0.733*** (0.119)
ln(age)	1.599*** (0.203)	1.546*** (0.200)	1.556*** (0.200)	2.041*** (0.225)
ln(age) ²	-0.235*** (0.029)	-0.224*** (0.029)	-0.226*** (0.029)	-0.271*** (0.031)
Crop / plant products		0.122 (0.094)	-0.052 (0.106)	-0.011 (0.108)
Wild harvest		0.712*** (0.230)	0.521** (0.229)	0.586** (0.232)
Processed products		-0.553*** (0.123)	-0.622*** (0.126)	-0.575*** (0.127)
Organic baskets		-0.943 (0.704)	-1.371* (0.734)	-1.399* (0.731)
Alcoholic beverages		-0.459 (0.319)	-0.531* (0.315)	-0.566* (0.313)
Observations	430,076	430,076	423,362	423,362
Enterprises	4,114	4,114	4,080	4,080
Log-likelihood	-4920.44	-4899.72	-4857.93	-4805.32

Notes: This table shows cloglog estimates of the monthly exit hazard on the long-horizon patience proxy. Dependent variable: 1 = complete exit at month t . Standard errors clustered at the enterprise level in parentheses. Columns (3)–(4) include certifier and region fixed effects (coefficients not reported). Column (4) adds entry-cohort fixed effects. Long-horizon proxy = 1 if certified for maple/forest products or livestock. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure D.2 confirms the pattern non-parametrically: the Kaplan–Meier survival curve for long-horizon enterprises lies strictly above that of short-horizon enterprises throughout the observed lifecycle, and a log-rank test rejects equality of the two curves. The negative relationship between the long-horizon proxy and exit hazard supports the view that preference heterogeneity drives observed exit timing: producers who undertake investments that only pay off on multi-year horizons are precisely those the model identifies as patient, and they exit at substantially older ages. This evidence does not uniquely identify the structural discount factors, but it confirms that the model’s central mechanism is consistent with observable cross-sectional variation in exit timing.

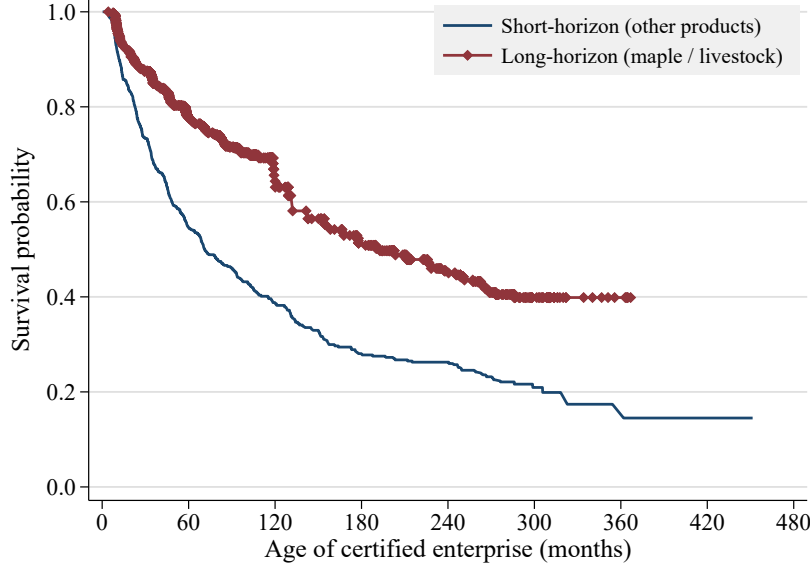


Figure D.2: Kaplan–Meier Survival by Long-Horizon Certification Proxy

Notes: This figure shows Kaplan–Meier survival curves by patience proxy group. The navy curve corresponds to short-horizon enterprises (all product categories other than maple/forest or livestock); the maroon curve with diamond markers corresponds to long-horizon enterprises certified for maple syrup / forest products or livestock. Shaded bands: 95% pointwise confidence intervals. Log-rank test rejects equality of survival distributions ($\chi^2(1) = 86.82, p < 0.001$).

D.2 Sensitivity Analysis

The structural analysis consists of two complementary exercises: structural restriction tests that assess whether the model’s key mechanisms are necessary to generate the observed hazard profile, and a parameter sensitivity grid that assesses quantitative robustness.

Structural Restriction Tests. Table D.4 and Figure D.3 report the model’s hazard profile under four structural restrictions. The first set replaces the two calibrated discount factors with a single common value, eliminating preference heterogeneity while keeping all other parameters at $\hat{\chi}$. Under any homogeneous patience assumption, whether all-impatient ($\beta = \hat{\beta}_\theta$), average ($\beta = \hat{\beta}$), or all-patient ($\beta = \hat{\beta}_{\bar{\theta}}$), the hazard rises continuously throughout the horizon without ever declining: there is no inverted-U. The intuition is that without compositional selection, the share of impatient producers never falls, so the aggregate hazard cannot turn downward. This result confirms that heterogeneous patience is not merely consistent with the hump-shaped hazard; it is necessary to generate it.

The second set keeps discount factors at $\hat{\chi}$ but replaces $\hat{\Omega}$ with two alternatives. The first imposes i.i.d. pest shocks by setting every row of Ω equal to the long-run distribution of pest states implied by $\hat{\Omega}$: the fraction of time the calibrated Markov chain spends in each state in the long run. Formally, $\hat{\omega}$ is the unique probability vector satisfying

$$\hat{\omega}^\top \hat{\Omega} = \hat{\omega}^\top, \quad \sum_j \hat{\omega}_j = 1. \quad (\text{D.1})$$

Because $\hat{\Omega}$ assigns high persistence to the worst pest state ($\hat{\Omega}_{33} = 0.85$), the chain spends most of its time

there, giving

$$\hat{\omega} = (0.046, 0.196, 0.758). \quad (\text{D.2})$$

Setting every row of Ω to $\hat{\omega}^\top$ yields

$$\Omega^{\text{iid}} = \begin{pmatrix} 0.046 & 0.196 & 0.758 \\ 0.046 & 0.196 & 0.758 \\ 0.046 & 0.196 & 0.758 \end{pmatrix}, \quad (\text{D.3})$$

so that the probability of reaching any given state is independent of the current state, eliminating shock persistence while preserving the long-run distribution of pest states implied by $\hat{\Omega}$. The second is absorbing states ($\Omega = \mathbb{I}_3$), where each enterprise is locked into its initial pest state forever. Under i.i.d. transitions the inverted-U shape is preserved but its peak shifts earlier, because without state persistence the selection mechanism operates more abruptly. Under absorbing states, the third of enterprises stuck in the severe pest state at certification exit en masse in the first period while those initially in favourable states essentially never exit. Long-run survival converges to roughly two-thirds of the entry cohort, the share initially drawn from the two favourable states, and the inverted-U is eliminated because there is no interior peak: all exit risk is concentrated in the first period. Pest-shock persistence, not merely pest-shock severity, is therefore also a necessary ingredient of the mechanism.

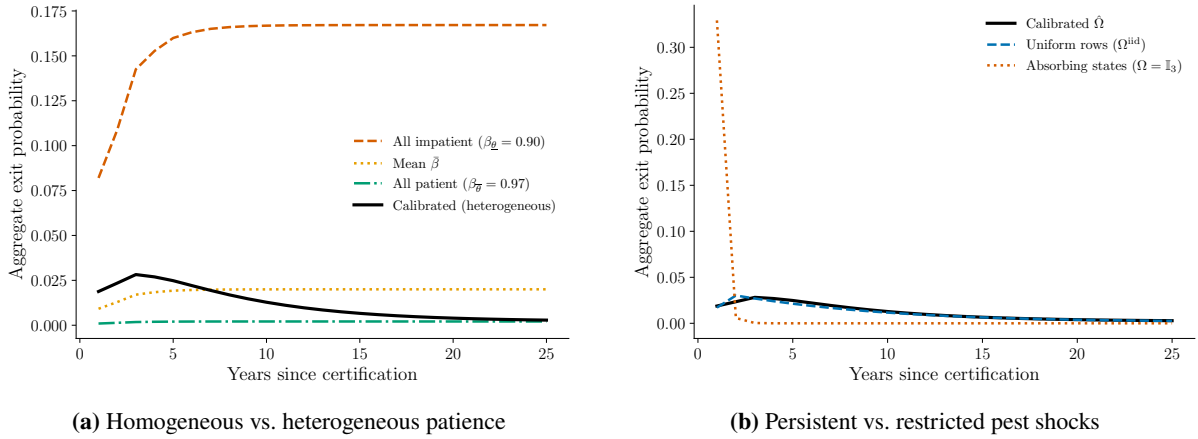


Figure D.3: Structural Restriction Tests: Hazard Profiles

Notes: This figure shows the aggregate annual exit probability P_t under four structural restrictions of the calibrated model. Left panel: calibrated model (solid black) against three homogeneous-patience restrictions in which both discount factors are set to a common value. Right panel: calibrated model against two restrictions on the pest-shock transition matrix: uniform rows (Ω^{iid} , each row equal to $(0.046, 0.196, 0.758)$ so the next state is independent of the current state) and absorbing states ($\Omega = \mathbb{I}_3$, each enterprise locked into its initial pest state). Crop-plant sub-sample, $n_0 = 1,683$.

Table D.4: Structural Restriction Tests: Necessity of Heterogeneous Patience and Persistent Pest Shocks

Test	Variant	Peak year	S_{10}	S_{20}	Inverted-U
<i>Discount factor restrictions (calibrated $\hat{\Omega}$ maintained)</i>					
	All impatient ($\beta_{\underline{\theta}}$)	–	0.202	0.033	×
	Mean $\bar{\beta}$	–	0.837	0.684	×
	All patient ($\beta_{\bar{\theta}}$)	–	0.982	0.961	×
	Calibrated (heterogeneous β)	3	0.810	0.757	✓
<i>Transition matrix restrictions (calibrated $\beta_{\underline{\theta}}, \beta_{\bar{\theta}}$ maintained)</i>					
	Calibrated $\hat{\Omega}$	3	0.810	0.757	✓
	Uniform rows (Ω^{iid})	2	0.821	0.769	✓
	Absorbing states ($\Omega = \mathbb{I}_3$)	–	0.667	0.667	×

Notes: This table presents structural restriction tests assessing whether heterogeneous patience and persistent pest shocks are necessary to generate the observed hazard profile. Discount factor restrictions replace both $\hat{\beta}_{\underline{\theta}}$ and $\hat{\beta}_{\bar{\theta}}$ with a single common value, eliminating preference heterogeneity while holding all other parameters at their calibrated values. Mean $\bar{\beta} = \hat{S}_0^{\underline{\theta}} \hat{\beta}_{\underline{\theta}} + (1 - \hat{S}_0^{\underline{\theta}}) \hat{\beta}_{\bar{\theta}}$. Transition matrix restrictions replace $\hat{\Omega}$ with uniform rows (Ω^{iid} , see text) or absorbing states ($\Omega = \mathbb{I}_3$). S_{10} : ten-year aggregate survival. S_{20} : twenty-year aggregate survival.

Parameter Sensitivity. The model’s qualitative predictions are robust throughout the empirically plausible parameter space. Table D.5 perturbs pest persistence (Ω_{33}) and the organic profit (π_o) by $\pm 10\%$ and $\pm 20\%$ of their calibrated values (with Ω_{33} capped at its admissible upper bound 0.99), and the discount factors ($\beta_{\underline{\theta}}, \beta_{\bar{\theta}}$) over $\beta_{\underline{\theta}} \in [0.87, 0.93]$ and $\beta_{\bar{\theta}} \in [0.93, 0.99]$ around the calibrated values $\hat{\beta}_{\underline{\theta}} = 0.90$ and $\hat{\beta}_{\bar{\theta}} = 0.97$ (with $\beta_{\bar{\theta}}$ capped at the admissible upper bound 0.99); the last row varies all three groups jointly across all 125 combinations. Within the inner perturbation grid (27 combinations), the inverted-U hazard profile is preserved in 22 cases. The qualitative policy ranking (the pesticide tax generates more durable long-run retention than a one-year subsidy) holds across all 125 grid points. The 55 exceptions to the inverted-U (5 in the inner grid, 50 at the boundaries) split cleanly into degenerate regimes: 42 rapid-exit cases ($S_{10} < 0.05$), driven by high pest persistence or by depressed profit, where survival collapses before a peak can form; 9 saturation cases ($S_{10} > 0.95$), driven by the high-profit/high-patience corner, where survival saturates near one and the hazard stays flat; and 4 intermediate boundary cases that fail the peak-year filter.

Table D.5: Sensitivity Analysis: Quantitative Robustness of the Calibrated Model

Perturbation	Range	Min S_{10}	Max S_{10}	Peak year range
Discount factors ($\beta_{\underline{\theta}}, \beta_{\bar{\theta}}$)	$\beta_{\underline{\theta}} \in [0.87, 0.93], \beta_{\bar{\theta}} \in [0.93, 0.99]$	0.422	0.892	3–5
Pest persistence (Ω_{33})	$\Omega_{33} \in [0.680, 0.990]$	0.000	0.982	1–5
Profit differential (π_o)	$\pi_o \in [80\%, 120\%] \times \hat{\pi}_o$	0.000	0.977	1–7
All perturbations jointly	125 combinations	0.000	1.000	1–25

Notes: Each row varies one dimension at a time over $\pm 10\%$ and $\pm 20\%$ of the calibrated value (last row: all three jointly). Across all 125 grid points, the inverted-U hazard profile is preserved in 70 cases (56.0%) and the qualitative policy ranking (pesticide tax generates more durable long-run retention than a one-year subsidy) is preserved in 125 cases (100.0%).